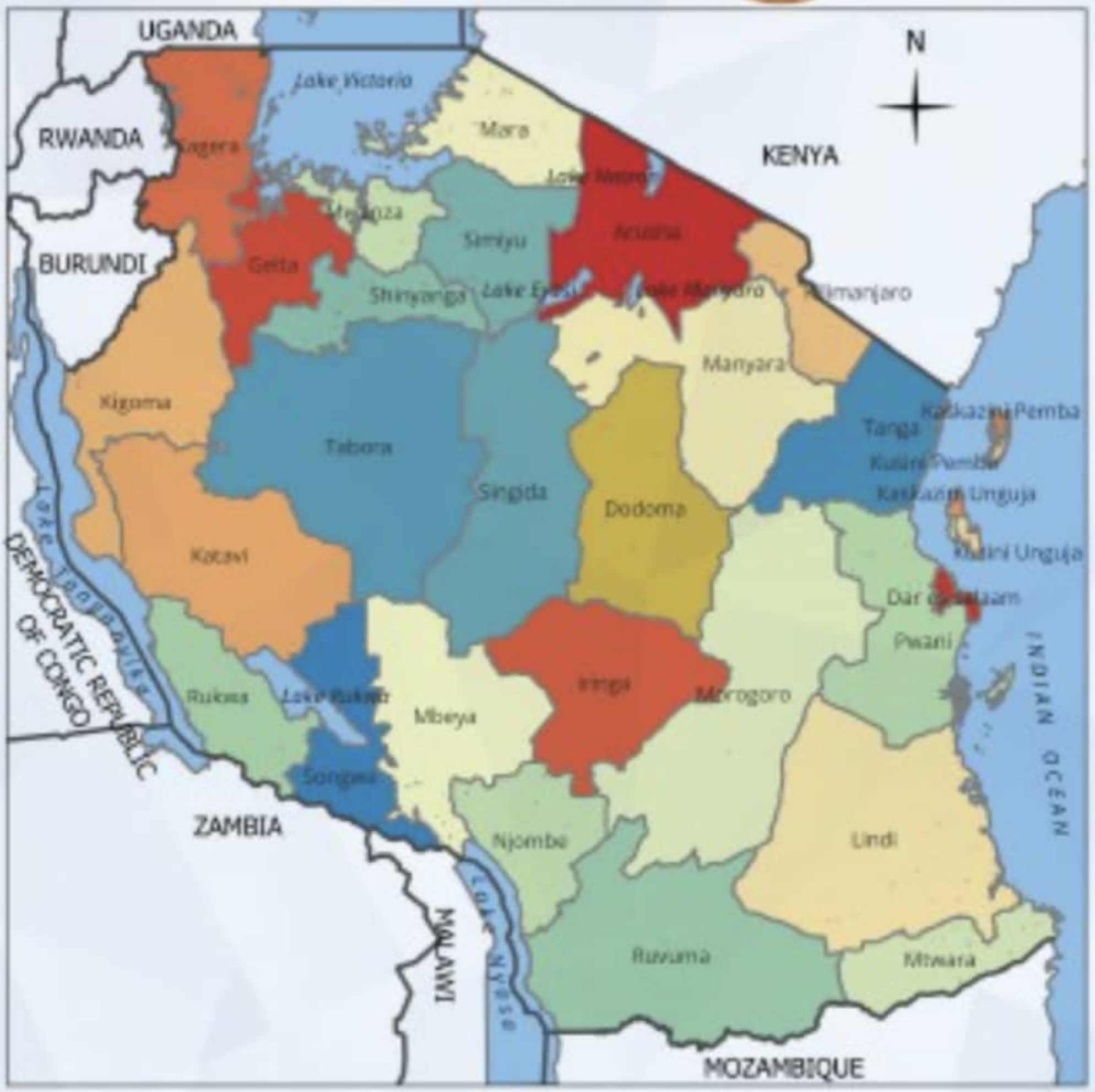
Geography and the Environment

Pupil's Book

Standard 4



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Geography and the Environment

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Standard 4

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MINISTRY OF EDUCATION,

SCIENCE AND TECHNOLOGY

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Dr Lyabwene M. Mtahabwa

Commissioner for Education

Tanzania Institute of Education

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Dr Aneth A. Komba
Director General
Tanzania Institute of Education

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Introduction

This textbook, Geography and the Environment for Primary Schools is written based on the Geography and the Environment syllabus for primary education standards III-VI, issued by the Ministry of Education, Science and Technology in 2023. Some of the contents in this book have been taken from the Social Studies textbooks for Standard III, IV, and V, published in 2018, following the 2016 syllabus. This textbook contains four chapters: The concept of a map, the Cardinal Directions, Drawing a simple map, and The uses of maps. By learning the contents of this textbook, you will develop the knowledge, skills, and attitude necessary for map drawing, reading, and interpretation.

The content of each chapter is presented in the form of text, activities, different maps and diagrams to enable you to develop the intended competencies. Each chapter has sufficient exercises to test your understanding of the specific themes. Therefore, you are advised to do all the exercises and activities in this textbook, along with any additional activities provided by the teacher to develop the intended competencies.

Additional learning resources are available in the TIE library, at *https://ol.tie.go.tz* or *ol.tie.go.tz*

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Chapter One

The Concept of a map

Introduction

The landscape of the earth's surface varies from one area to another. Normally, it is difficult to understand the landscape of a place without visiting a specific location. The map is a guide that simplifies learning about a certain area on the earth's surface without having a physical visit to the place. In this chapter, you will learn about the meaning of a map, the types of maps, the essential elements of maps, the characteristics of a good map, and the importance of maps. The competencies developed will enable you to elaborate types of the maps in different contexts.



Think

The layout of different places on the earth's surface

Meaning of a map

A map is a drawing that represents the information found on the earth's surface by using a specific scale. Maps can be drawn on a paper, a piece of cloth, digital, or any flat area. The geographical information that can be represented on a map includes mountains, valleys, rivers, roads, buildings, and other important information. Maps help us to understand places and how to access them.

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Exercise 1

1. Why do we need maps?
2. Explain the difference between a picture and a map.

Types of maps

Different criteria are used to identify the types of maps. One of the criteria is the purpose of the map. By using this criterion, there are two types of maps namely; general maps and thematic maps.

General maps

These are maps that provide general information about a certain area. These maps do not present just one piece of information but offer various details about the area. For example: they can include information about roads, rivers, administrative boundaries, settlements, mountains, and vegetation. These maps help the user to understand the basic information about a place, the relationship between different features presented on the map, and to identify directions on the earth's surface. Therefore, general maps include: topographical maps, political maps, and road and railway maps.

Topographical maps

These maps represent the layout of natural and man-made features on the earth's surface. Also, topographical map use contour line to show the terrain of an area. Example of natural features include: ocean, lakes, rivers, plains, valleys, mountains, and vegetation. Example of man-made features include buildings, and roads, as shown in Figure 1. These maps are important in planning for construction, tourism, farming, industries, fishing, and business activities.

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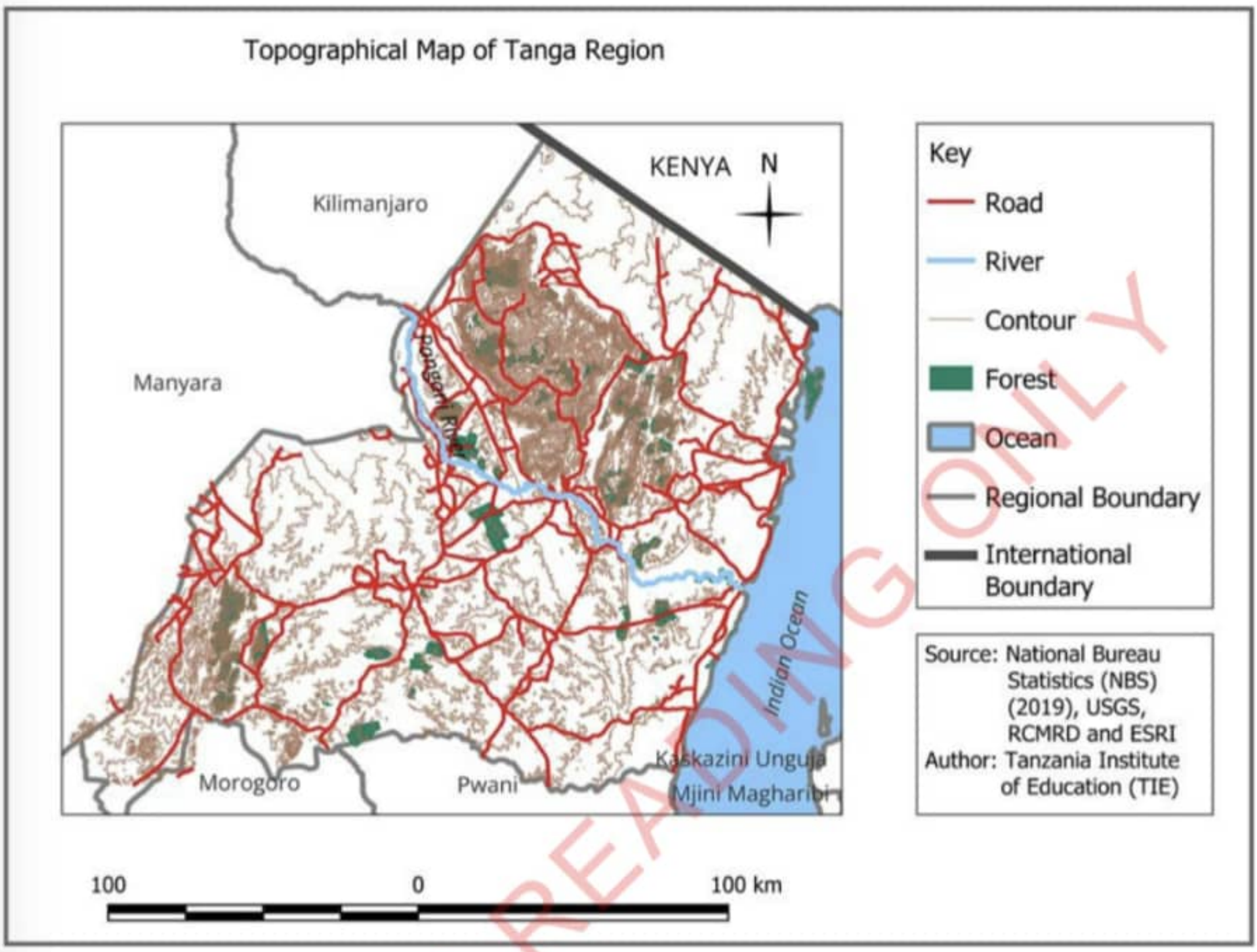


Figure 1: *Topographical map of Tanga region*

Political maps

These maps provide information about the international, regional, district and other local boundaries. They also show important areas such as government buildings, capital cities, and towns, as shown in Figure 2. Political maps are helpful for researchers, learners, and visitors who need to understand the administrative boundaries of a certain area.

Political maps change over time. Several factors can cause changes in a political map. These include wars, boundaries conflicts between countries, the nationalism, political agreements, and a boundary changes within a country. Because of these factors, political maps are updated to show the current national boundaries or other specific areas.

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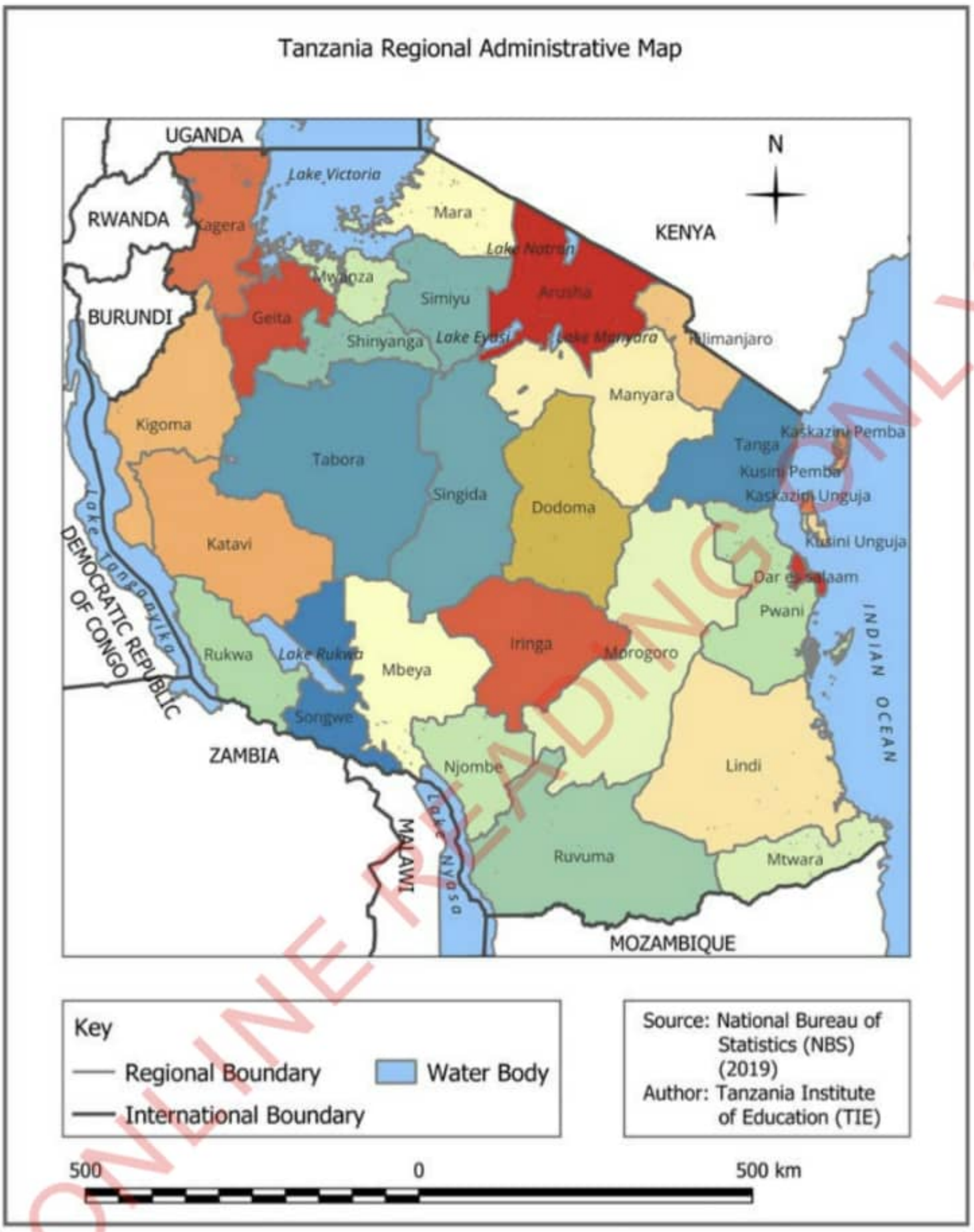


Figure 2: *Regional administrative boundaries*

Road and railway maps

These maps provide important information about road and railway networks, such as highways and street roads, and important signs or symbols of a certain area as shown in Figure 3. They also show the distance between different locations and often provide information about petrol stations, restaurants, and hotels. Road and railway maps are important for drivers and travellers.

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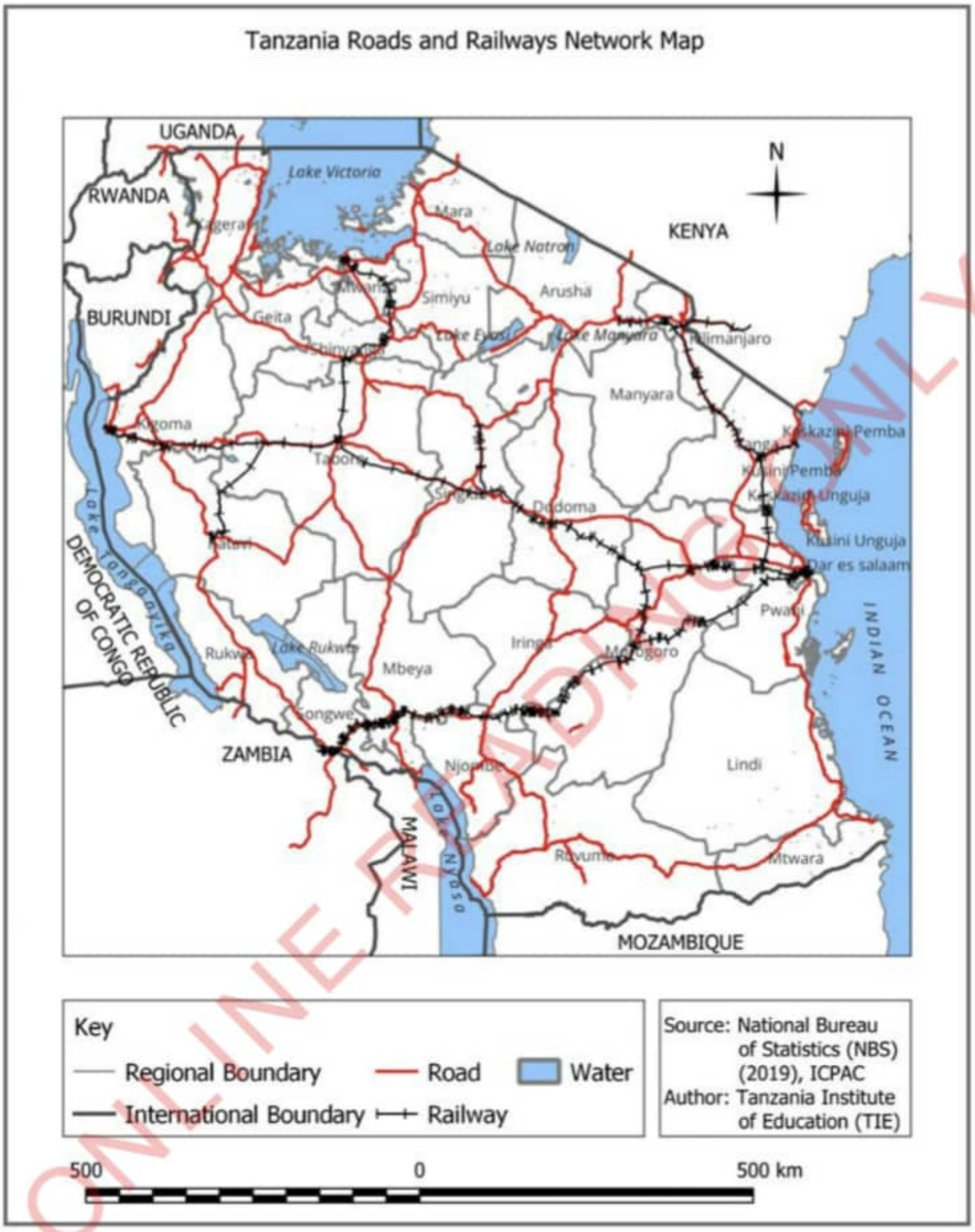


Figure 3: *Road and railway map*

Exercise 2

1. What type of map will you use to reach a region you have never been to?
2. Why do political maps change over time?
3. If you want to climb a mountain, what type of map would you use to understand the elevation of that mountain?

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Thematic maps

These maps represent specific and unique themes or information. Thematic maps can vary depending on the theme being presented. For example, thematic maps can represent types of soils, vegetation, crops, human populations, weather, and urban planning.

Population map

This map represents the density or distribution of people in a certain area. Population maps help us see where areas are crowded or not crowded. They are useful for people who make decisions about where to put important things like schools, hospitals, markets, and roads. Figure 4 shows a map of population distribution in Tabora Region.

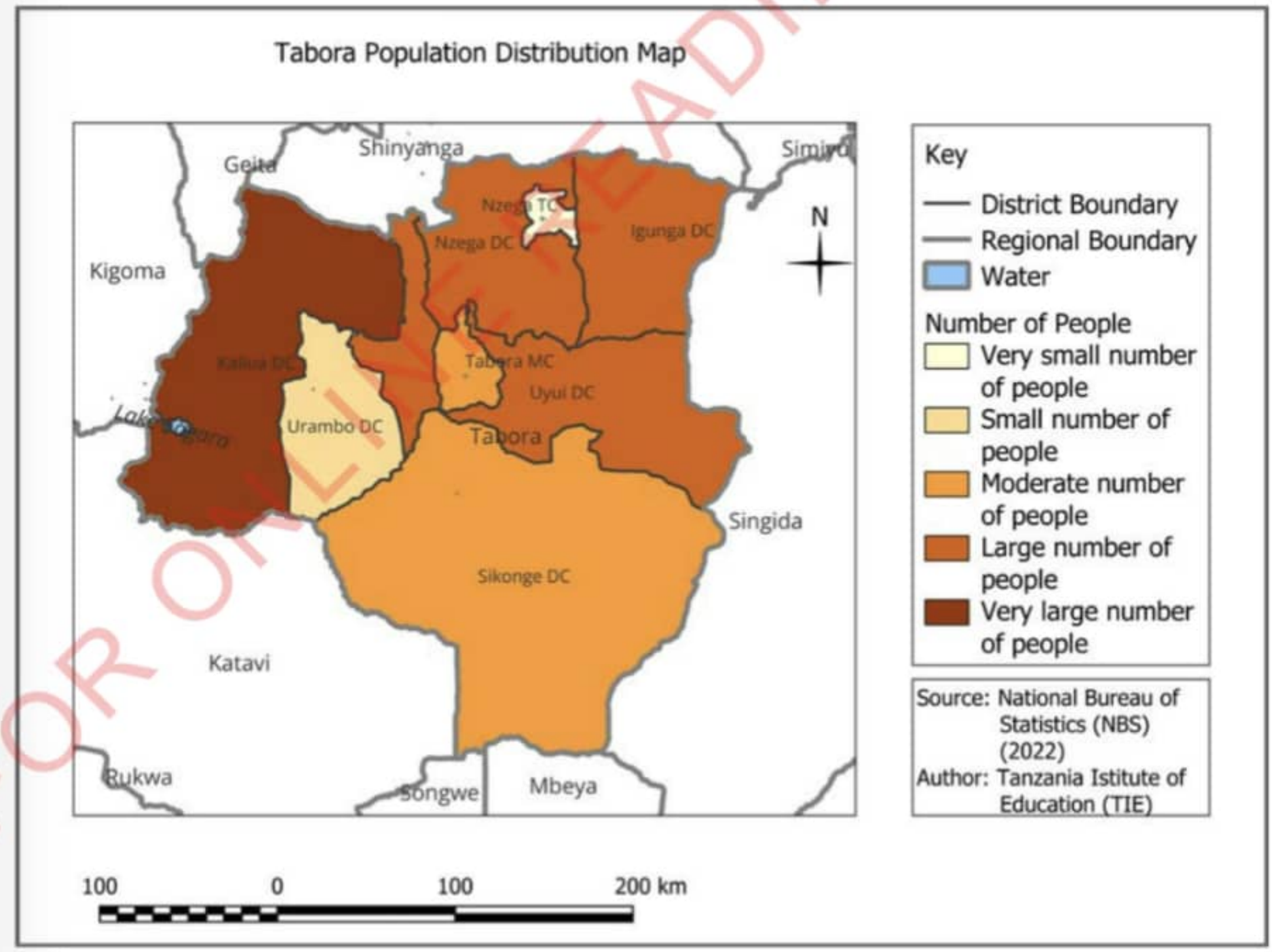


Figure 4: *Population distribution in Tabora Region*

Weather maps

These maps represent the distribution of weather in a certain area.

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They illustrate elements of weather such as temperature, rainfall, wind, humidity, and atmospheric pressure, and how they differ from one place to another. These maps are important for farming, business, tourism, and transportation activities. Figure 5 shows rainfall distribution in Iringa region.

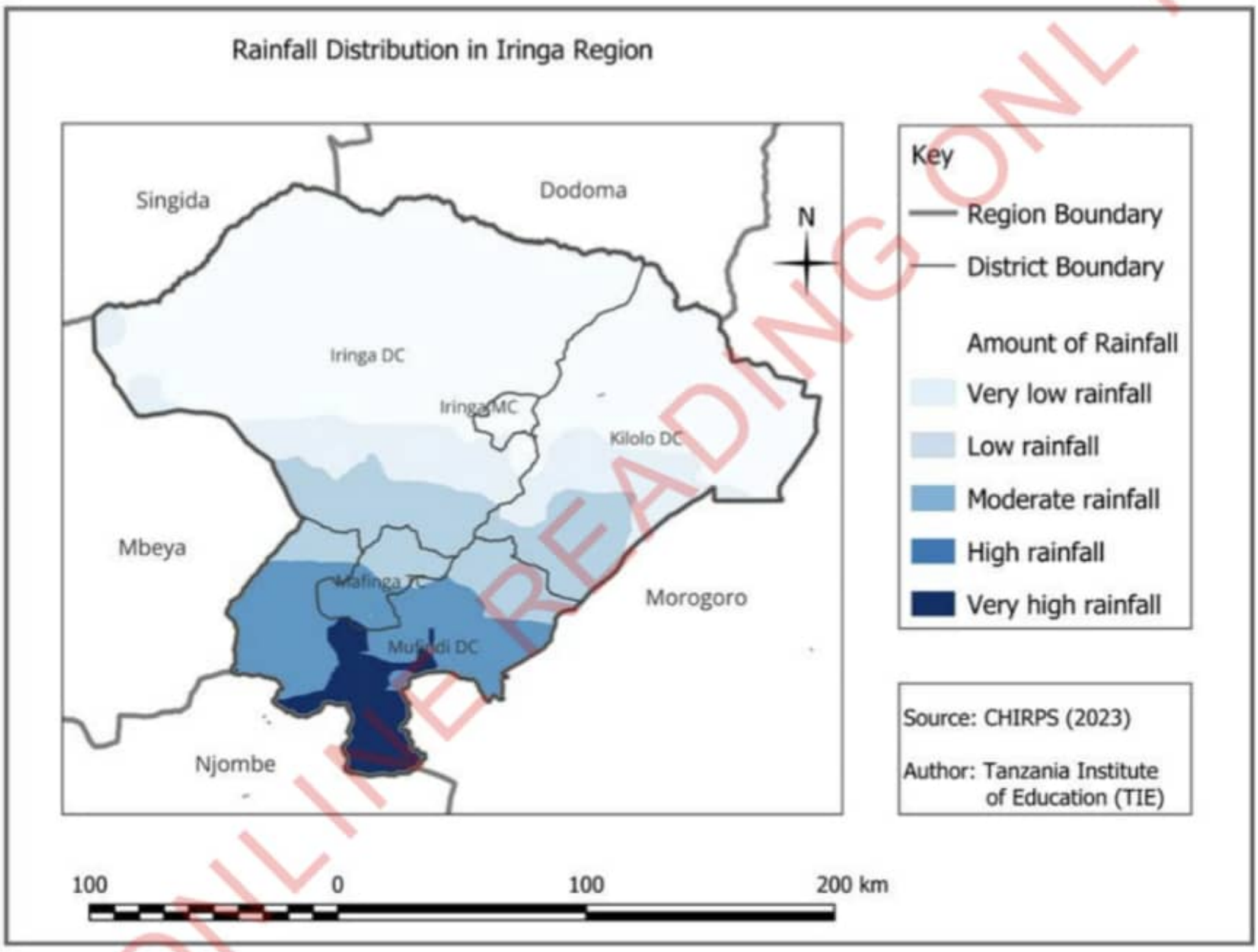


Figure 5: *Rainfall distribution in Iringa Region*

Urban planning maps

These maps show the planning of towns, cities and villages for settlement and social services. For example the placement of houses, shops, schools, hospitals, and roads as shown in Figure 6. These maps are mostly used by land surveyors and urban planning officers.

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Figure 6: *Urban planning map*

Exercise 3

1. Why thematic maps can greatly vary even if they show the same area?
2. If there is a fire emergency in your neighbourhood, what type of map would you use to direct the firefighters to the area?
3. Carefully observe the following table, then suggest an appropriate map that may be used by the following people.

No.	A person	Type of map
1.	A builder	
2.	A driver	
3.	A tourist	
4.	A Farmer	

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Activity 1

Observe Figure 7, then answer the following questions:

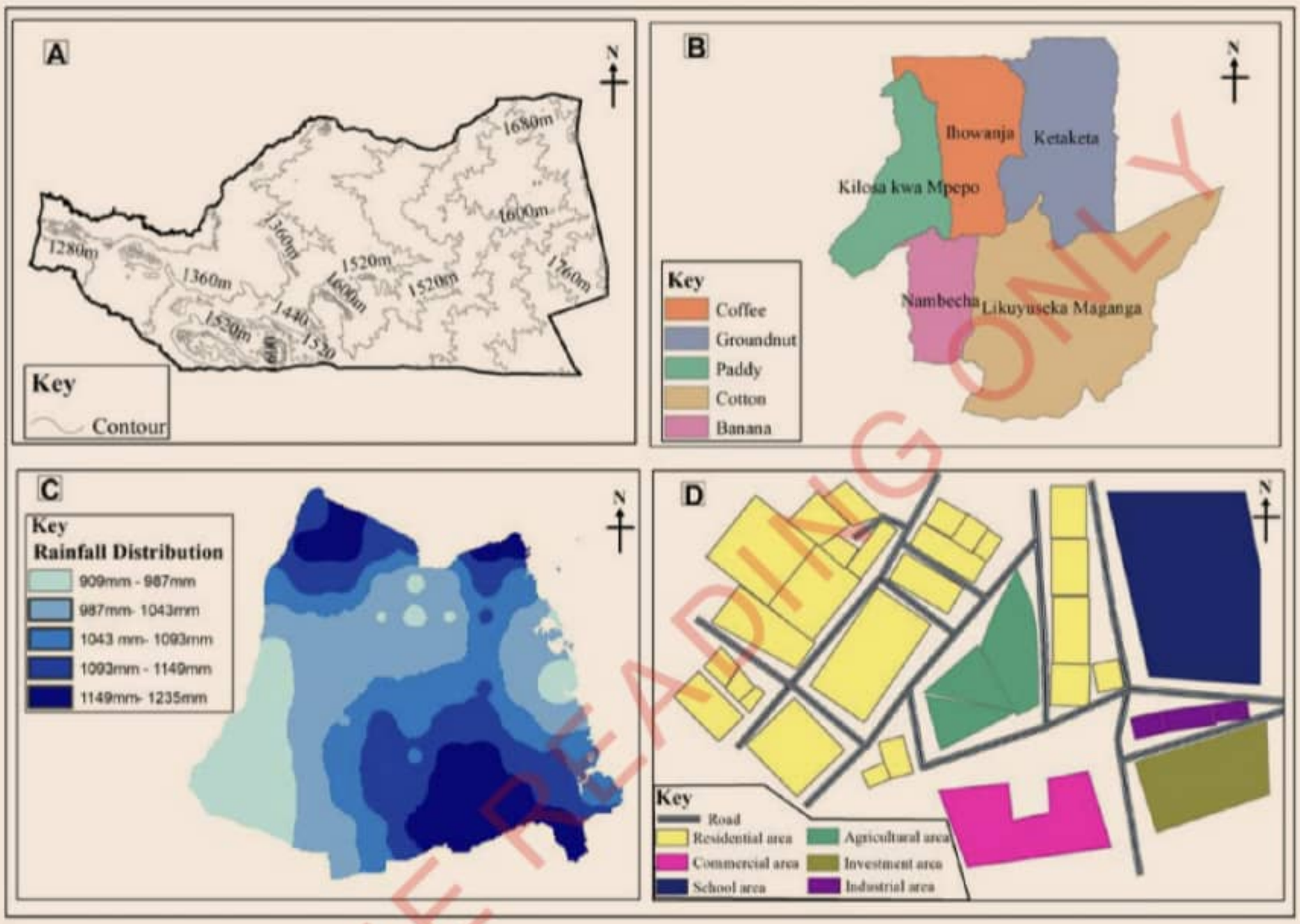


Figure 7: *Types of map*

Questions

1. Identify the type of maps in A, B, C, and D as presented in Figure 7.
2. Identify thematic maps and general maps shown in Figure 7.

Essential elements of a map

The essential elements of a map are the basic parts that help in the use of a specific map. These elements give the map its meaning and make it accurate. They help to make the map easier to understand and present information about the place correctly. Here are the essential elements that should be included on a map.

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(a) Frame: a border or outer line surrounding the map. The map frame helps to clearly define the map's boundaries and separates it from other information on the page. It shows where the map begins and ends;

(b) Title: helps to introduce a map to the users by identifies what a map is about or the geographical information or features it represents. For example, if the map shows roads in Tanzania, the title of the map might be 'Tanzania's roads map;

(c) North Direction: it indicates the direction on the map. It is usually represented by a North arrow. This element helps the reader to identify other directions such as South, East, and West. Knowing directions on a map helps you to understand where things are located;

(d) Key: it explains the signs and symbols used on a map. For example, a symbol of a tree might represent a forest, and a blue wavy line might represent a river;

(e) Scale: it shows the relationship between the distance on a map and the real distance on the ground. For example, one centimetre on the map might represent ten kilometres on the ground;

(f) Grid Lines: these are lines that divide a map into squares. They help to locate specific areas on the map. Each square has a number or letter to help you find things easily; and

(g) Source: it shows where the information presented on a map was obtained. The source can be a government organisation, a non-government organisation, or an individual. Showing the source of the map helps verify its accuracy and allows other users to refer to it in their work. Figure 8 shows the essentials of a map.

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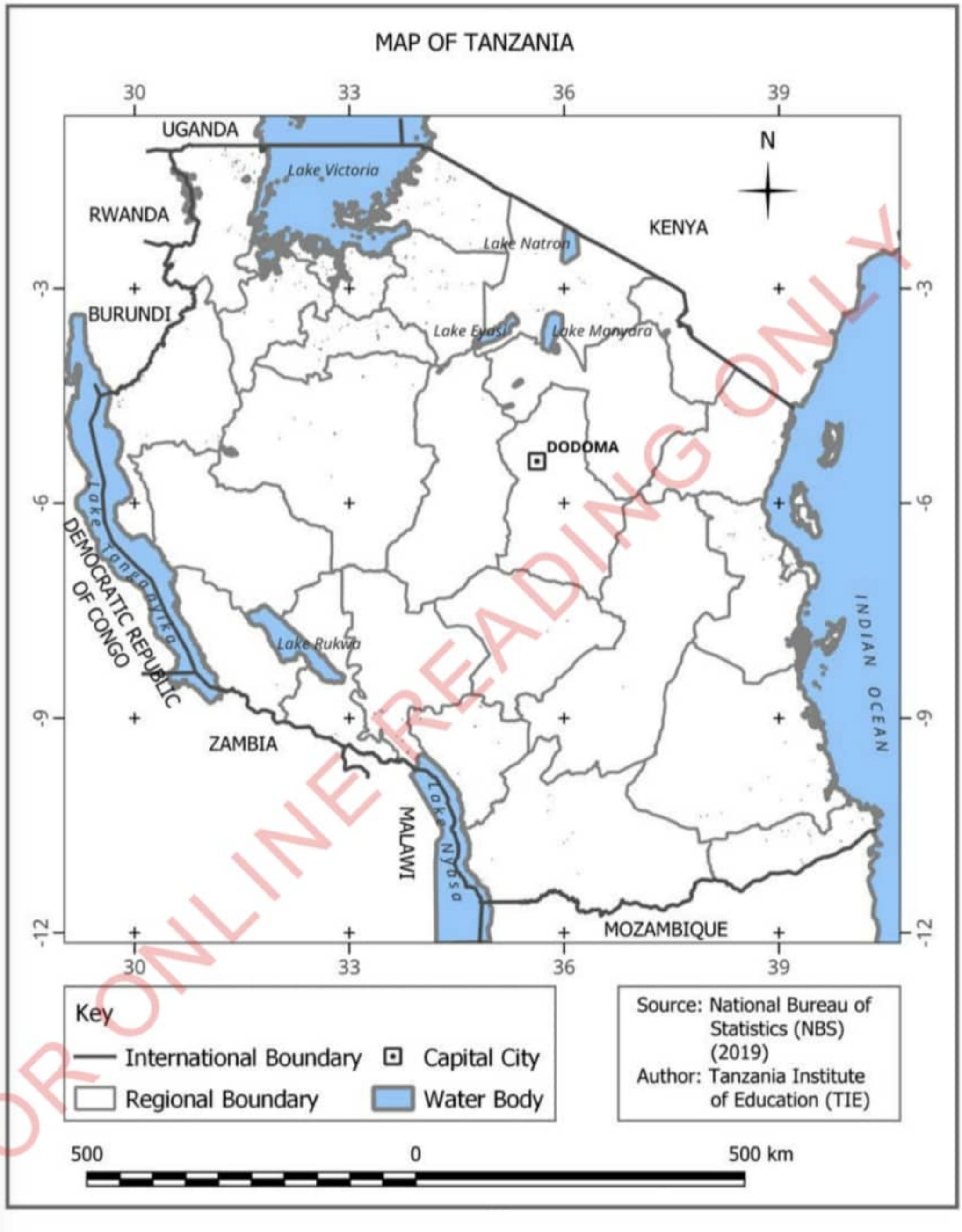


Figure 8: *Essential elements of a map*

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Activity 2

Observe Figure 9, then answer the following questions.



Figure 9: *Mwendapole Village*

Questions

1. Explain the weaknesses you have identified in Figure 9.
2. Identify the essentials of a map that should be placed at points A, B, C and D, then explain the importance of each element you have identified.
3. Identify the type of maps represented by Figure 9.

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Characteristics of good maps

A good map should have the following characteristics:

(a) It should be easy to read and interpret;

(b) it should include all the essential elements of a map, such as scale, frame, source, key, grid lines, title, and North direction;

(c) it should represent the actual places, and distances, as well as place objects correctly;

(d) it should use clear colours and neat writing to help users understand it better;

(e) it should only show the information that is needed for specific purpose;

(f) it should use the same symbols, colours, and styles consistently across the map for easy understanding; and

(g) it should have the up to date information so that everything shown is correct.

Importance of maps

A map is like a guidebook for the world, helping us learning by explore, and understand the places around us. Therefore, a map is an important tool used by people in various contexts. The following are some key reasons for using maps:

(a) To find and learn about places, even if we have never been there before;

(b) to know distances from one place to another;

(c) to know the right direction to take to reach our destination, such as North, South, East, or West;

(d) to plan our travels by showing the best roads, shortcuts, and interesting places we might see along the way;

(e) to understand different features in our environment, such as mountains, rivers, lakes, and forests;

(f) to help during emergencies like fires and floods by guiding

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rescue teams to the best routes to reach the area for helping people;

(g) to learn about the weather in different places by showing rainfall distribution, vegetation cover, and human activities;

(h) to show where important resources in our country are found, such as rivers, lakes, forests, and minerals;

(i) to show where people live and how many people are in different areas; and

(j) to know administrative boundaries, such as international, regional, district, and village boundaries;

(k) to show tourist attractions such as Mount Kilimanjaro, the Ngorongoro Crater, and wildlife reserves. These help in attracting tourists to visit these places; and

(l) to understand the history of a place by showing how it has changed over time.

Exercise 4

1. How do maps assist in promoting tourism activities?
2. How can a farmer use a map?

Revision exercise

Section: A: Choose the letter of the correct answer.

1. What is a map?

(a) Representation of features of the earth's surface using the North direction.

(b) Representation of features of the earth's surface using a scale.

(c) Representation of features of the earth's surface using a key.

(d) Representation of features of the earth's surface using grid lines.

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2. Which of these maps show natural and man-made features?
   (a) Topographical maps.
   (b) Political maps.
   (c) Economic maps.
   (d) Thematic maps.
3. Which of the following is one of the characteristics of a good map?
   (a) Having many colours.
   (b) Having many attractive symbols.
   (c) Being readable and easy to interpret.
   (d) Having many scientific terms.
4. What is the importance of maps to human life?
   (a) Gaining expertise.
   (b) Helping to show various symbols used in a map.
   (c) Showing where important national resources are located.
   (d) Demonstrating the ability to draw maps.
5. How can maps assist during emergencies?
   (a) Showing how a fire is spreading.
   (b) Indicating how disasters occur.
   (c) Helping rescue teams to determine the right direction to the area.
   (d) Assisting rescue teams in extinguishing the fire.
6. How can maps help resolve land conflicts?
   (a) By showing the distribution of people.
   (b) By showing administrative boundaries.
   (c) By showing the direction of places.
   (d) By showing distances between locations.

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7. How can maps be a source of government revenue?
   (a) Guiding to the tourist attractions.
   (b) Showing areas with modern roads.
   (c) Helping people access healthcare services.
   (d) Showing areas for celebrations and entertainment.

Section: B: Matching items

8. Match the items in column A with those in column B to produce a meaningful statement

No.	Column A	Column B
(i)	Topographical maps	(a) on paper, land, a piece of cloth, and a computer
(ii)	Frame	(b) represent the features of the Earth's surface
(iii)	Essential elements of a map	(c) the relationship between the distance shown on the map and the real ground distance
(iv)	Scale	(d) scale, frame, North direction, title, key, source, and gridlines
(v)	Surfaces where a map can be drawn	(e) shows the boundaries of a map
		(f) should be easy to read for map users
		(g) explains the symbols and signs used on the map

Section: C: Short answer items:

9. What are the qualities of a good map? List them by giving examples.
10. In what ways do topographical maps differ from urban plan maps?

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Vocabulary	
Compass	a tool used to show direction, with an arrow pointing North at all times
Consistency	maintaining the same symbols and colours that represent the same thing on a map
Landscape	the way the land looks
Man-made features	things built by people
Physical visit	going to see a place in person
Population density	how many people live in a certain area
Surveyor	a person who measures land to help decide where things like buildings and roads should be placed
Urban planning	how cities and towns are designed and arranged

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Chapter Two

The Cardinal Directions

Introduction

When human beings walk, they follow a certain direction on Earth to reach their intended destination. The Cardinal Directions are used to determine the correct direction. Geographically, there are specific names used to identify the Cardinal Directions. In this chapter, you will learn about the meaning of the Cardinal Directions, as well as how to draw and identify Cardinal Directions. You will also learn about intercardinal directions and the importance of using Cardinal Directions. The competencies developed will enable you to use the Cardinal Directions to determine location and directions.



Think

Determination of a route using the Cardinal Directions.

The meaning of Cardinal Directions

The Cardinal Directions are basic reference points that are used to indicate the direction of different places on the Earth. These directions are North, South, East, and West. Each direction represents a specific part of the Earth. The Cardinal Directions are represented by letters. That is, the letter N for North, the letter S for South, the letter E for East, and the letter W for West. These symbols are commonly found on maps, compasses, and various navigational tools to assist in pinpointing a specific location and direction to reach the particular

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location. They generally serve as the foundation for determining other directional references.

Activity 1

Follow these steps to draw the Cardinal Directions:

(a) Go outside the classroom in the morning;

(b) draw a circle where you are standing;

(c) identify the direction of the sunrise and face that side;

(d) walk three steps forward while facing the sunrise, stop and write the word East (E);

(e) go back to the circle, then walk three steps backwards while facing the sunrise and write the word "West" (W);

(f) return to the circle, then stop and face the direction of the Sun;

(g) turn to your right, walk three steps, and write the word "South" (S);

(h) go back to the circle, again face the sunrise. Turn left, walk three steps forward, and write the word "North." (N);

(i) take a stick and draw straight lines from each word you have written on the ground to the circle; and

(j) observe that drawing and copy it in your exercise book.

The diagram drawn should resemble an additional sign. This diagram represents the earth's Cardinal Directions, as shown in Figure 1.

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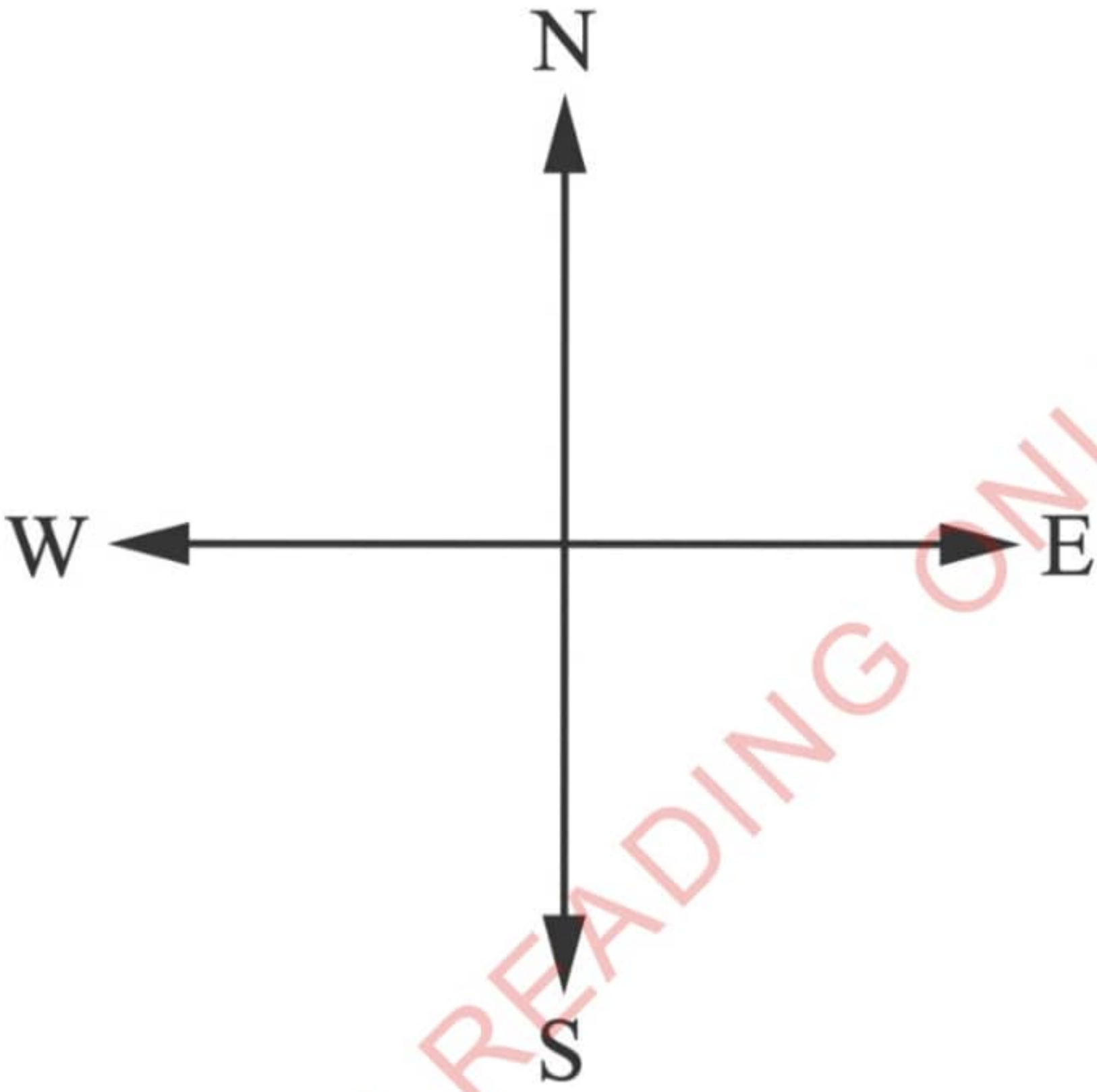


Figure 1: *The Cardinal Directions*

Exercise 1

Answer the following questions:

1. In which direction does the sunrise?
2. When you are facing the side where the sunrises, which cardinal direction will be to your left?
3. If you stand while your back is facing the direction where the sunrises, which direction of the Earth will be to:

(a) your left hand;

(b) your right hand; and

(c) in front of you.

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Methods of determining the Cardinal Directions

Cardinal Directions of the Earth can be identified using different methods. These methods include using sunrise and sunset, the direction of shadows, the direction of stars, maps, and compasses. At this stage, you will learn how to determine the Cardinal Directions by using sunrise and sunset, the direction of shadow, map and Compass.

Determining the Cardinal Directions using the sunrise and sunset

Watching the Sun as it rises in the morning or sets in the evening is one way to determine the Cardinal Directions in your location. Normally, the Sunrises in the East and sets in the West. This natural phenomenon helps you identify the East and West directions of your location. By understanding where the sunrises and sunsets, you can gain a basic understanding of the Cardinal Directions.

Therefore, in the morning, if you face the sun, you are facing East, and the West will be at your back. To your right side will be South, and to your left side will be North. Similarly, in the evening, if you face the sunset, you are facing the West, and the East will be at your back. To your right side will be North, and to your left side will be South.

Determining the Cardinal Directions using the shadow

The direction of your shadow in the morning and evening can be used to determine the Cardinal Directions. Usually, when you walk in the morning, the direction of your shadow falls to the West. For example, if you walk while facing the Sun, your shadow will be behind you, and if you walk while the Sun is to your back, your shadow will be in front of you, thus the direction will be West. Similarly, in the evening, the direction of your shadow will be East, as shown in Figure 2.

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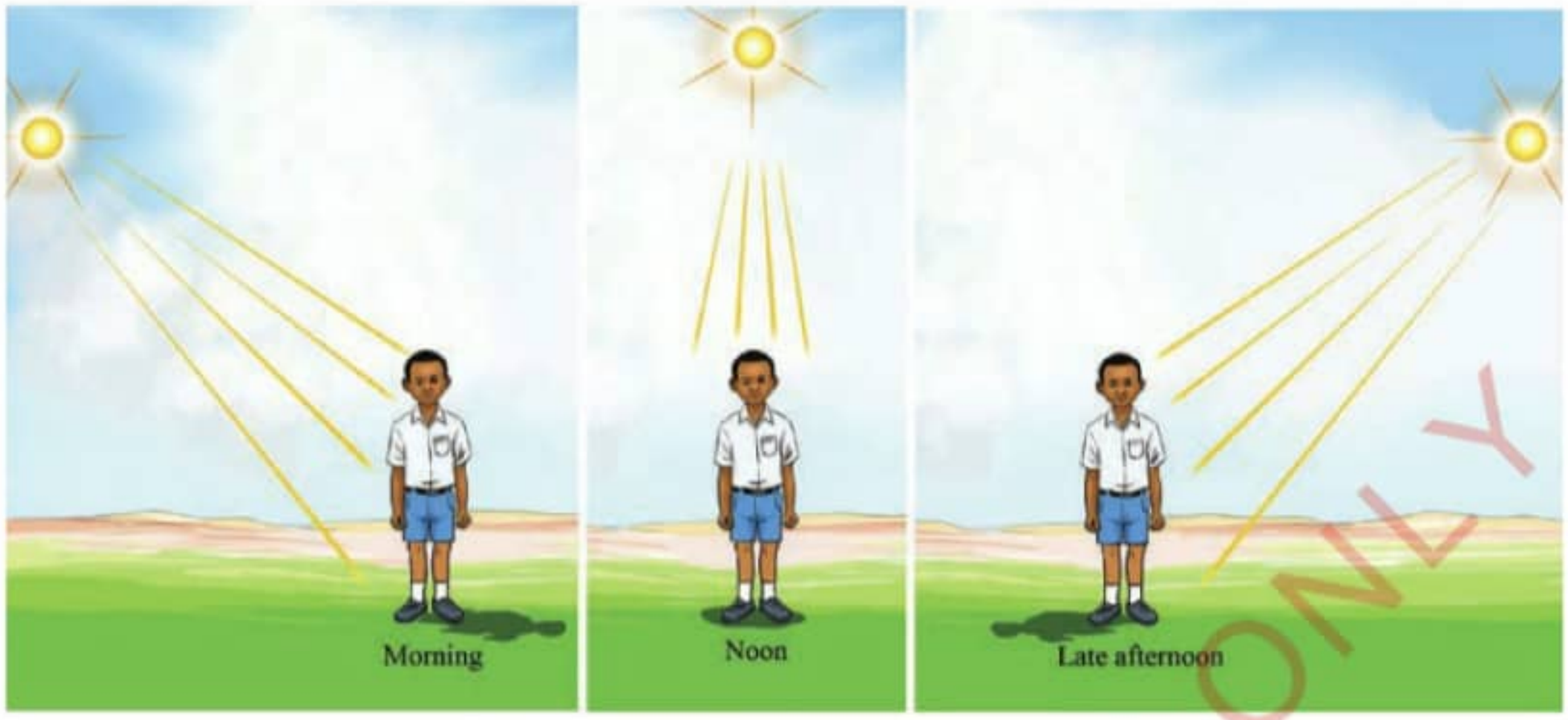


Figure 2: *Determining Cardinal Directions through the direction of shadow*

Note: It is impossible to identify the Cardinal Directions in the midday sun (when the sun is overhead). This is because, during this time, your shadow will be directly around you.

Activity 2

1. Go outside the classroom in the morning, and stand at the centre of your school compound.
2. Note the direction where the Sunrises, then identify the Cardinal Directions while naming the objects found in that direction.
3. Make a model of the Cardinal Directions using materials found in your surroundings.

Exercise 2

Answer the following questions:

1. What direction is behind you when you are facing North?
2. If your teacher is facing North, which direction will be on his / her left side?
3. If you are lost in the forest and notice that the sun is setting on your right side, which direction will you be facing?

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4. When you walk in the evening and your shadow is in front of you, which direction will you be facing?

Determining Cardinal Directions using maps

A map is one of the guides used to determine the Cardinal Directions. The North direction symbol is among the essential elements on a map. It used to simplify the determination of Cardinal Directions on a map namely: North, South, East, and West.

Activity 3

Observe Figure 3, then identify the direction of the objects according to the given instructions.

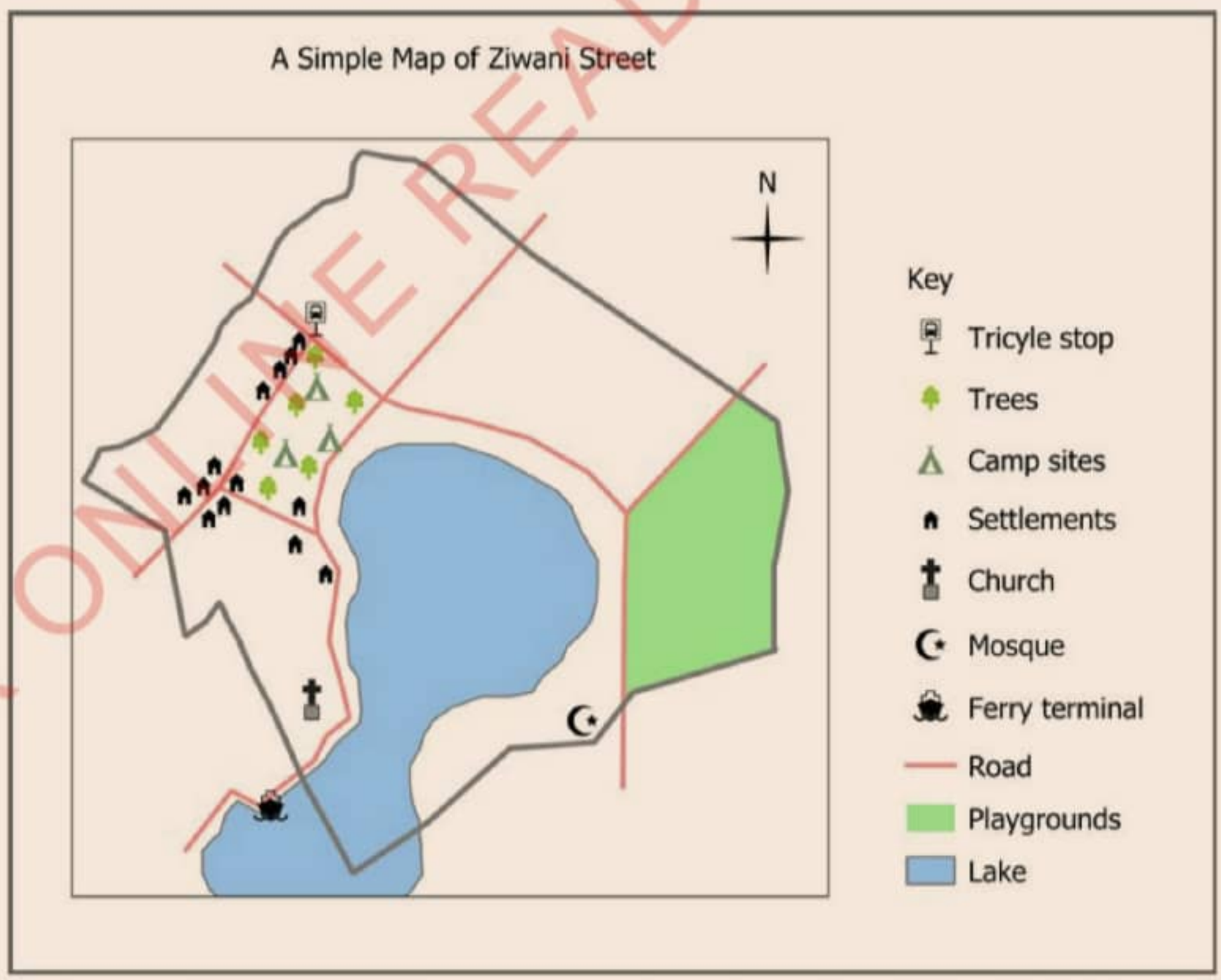


Figure 3: *Ziwani street*

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Questions

(a) Identify the direction of the tricycle stop from the campsite.

(b) Imagine, you are in a settlement area and you want to go to the ferry terminal, which direction will you follow?

(c) What is the direction of the Lake from the settlements?

(d) What is the direction of the Church from the campsites?

(e) What is the direction of the Mosque from the playground?

Determining the Cardinal Directions through a compass

A compass is a device used to identify the North direction. There are two types of compasses namely: digital compass and analog compass. Digital compass uses technology to detect and display the North direction. This type of compass is often integrated into GPS, modern watches, smartphones, and transport vehicles such as cars, boats, and planes as shown in Figure 4. An Analog compass uses a magnetized needle to point to the North.



Figure 4: *A digital compass*

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Steps to determine Cardinal Directions using a digital compass

(a) take your compass or download the compass App;

(b) click the App and and make sure the compass direction arrow is pointing the North (N);

(c) when the compass direction arrow is pointing North, orient yourself to that direction, this help you to detarmine the North direction of your location; and

(d) once you identify the North direction, you can determine the other directions. For example, the South will be at your back, the East will be on your right, and the West will be on your left.



Activity 4

Use a smartphone, tablet, or computer to download the Compass application (App), then use it to identify the Cardinal Directions of your location.

The intercardinal directions

The intercardinal directions are points that lie between the Cardinal Directions. They provide for more specific directional guidance to basic Cardinal Directions. These directions are Northeast (NE), Southeast (SE), Southwest (SW), and Northwest (NW). They offer more detailed explanations or specific directions, leading to a more precise and accurate understanding of a location. Intercardinal directions are commonly used in various fields, such as aviation, maritime navigation, geography, and weather forecasting.

Steps to draw the intercardinal directions;

(a) Draw the Cardinal Directions to show North, South, East, and West,

(b) draw a straight line between the North and East sides, starting from where those lines meet, then write Northeast (NE);

(c) draw another straight line between the South and East sides,

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then write Southeast (SE);

(d) draw a straight line between the South and West sides, then write Southwest (SW); and

(e) draw another straight line between the North and West sides, then write Northwest (NW).

After drawing the diagram, you will have eight directions. Those are intercardinal directions.

Note: When locating or reading the intercardinal directions, you should start with North or South. This is because these directions are the basic points on a compass used for finding directions, and they help you identify all other directions. Also, these direction are the basic guidance for identifying the intercardinal directions, as shown in Figure 5.

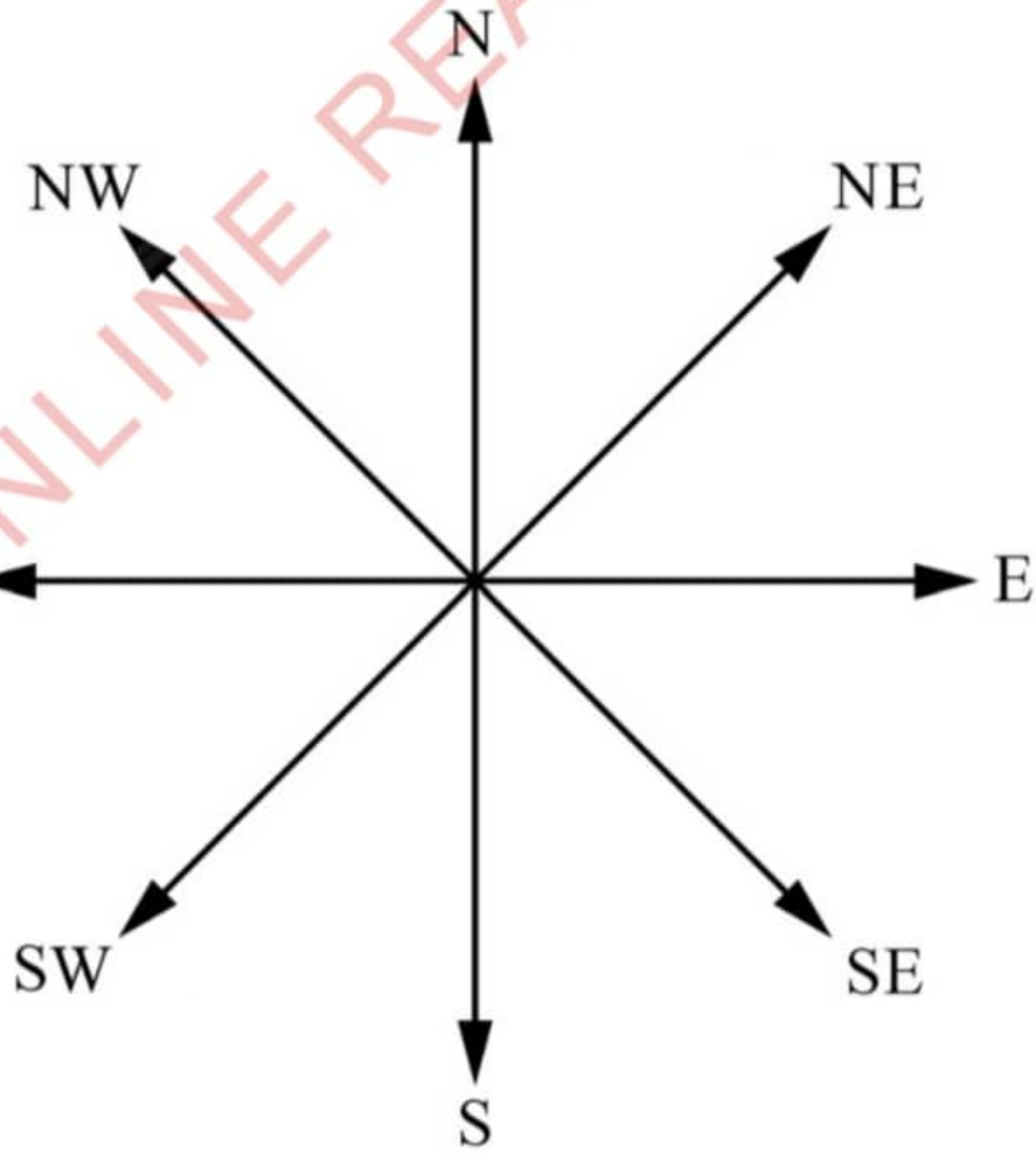


Figure 5: *The intercardinal directions*

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Activity 5

Observe Figure 6, then identify Sakina's direction in the provided table below:



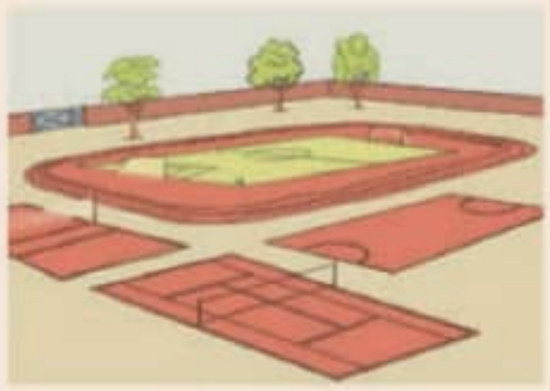
Library



Hotel



Dispensary


Play grounds




Dam


School


Home


Market

Figure 6: *Identifying the direction*

No.	Sakina's journey	Direction
(a)	Sakina is going to school	
(b)	Sakina is going to the library	
(c)	Sakina is going home	
(d)	Sakina is going to the market	
(e)	Sakina is going to the dam	
(f)	Sakina is going to the playgrounds	

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(g)	Sakina is going to the hotel	
(h)	Sakina is going to the dispensary	

Exercise 3

Answer the following questions:

1. Explain the differences between Cardinal Directions and intercardinal directions.
2. Why do we start with the North or South direction when identifying the intercardinal directions?
3. What direction is between South and East?
4. If your school is located Northwest of your friend's school, what is the direction of your friend's school?

The importance of Cardinal Directions

The Cardinal Directions are very important for daily human life. The following is the importance of the cardinal direction.

(a) It helps to identify directions and places on maps and the earth's surface. This makes it easier for users to identify their present location and where they are heading. They also help to direct people to reach their destination easily;

(b) it is useful to engineers and architects in designing buildings, roads, and railways. Understanding Cardinal Directions is crucial for determining the orientation of a construction site, taking into account factors like the direction of the sunrise and sunset. It also aids in planning the direction of roads and railways from one location to another;

(c) it is useful in guiding the movement of transport vessels. The

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Cardinal Directions help pilots, drivers, and captains choose the shortest and safest route; and

(d) it is useful for safety and security agencies. The Cardinal Directions help safety and security agencies in communicating and giving instructions during dangers and emergencies.



Activity 6

At school or home, play a hide-and-seek game with your friends. The seeker should then follow the directional instructions given by the game leader to find those who are hiding.

Revision exercise

Choose the letter of the correct answer

1. What is the direction of your left side when facing the sunset?
   (a) West
   (b) South
   (c) Norh
   (d) East
2. What is the abbreviation for the Southeast intercardinal direction?
   (a) SW
   (b) ES
   (c) SE
   (d) SN
3. If you are heading towards where sunset occurs, what direction are you facing?
   (a) North
   (b) South
   (c) East
   (d) West
4. Which direction is the opposite of the sunset?
   (a) North

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(b) South
(c) East
(d) West

5. Which direction does a compass arrow usually point?
   (a) North
   (b) South
   (c) East
   (d) West

Short answer questions

6. Why is an understanding of the Cardinal Directions important?
7. Which direction is located between North and East?
8. Observe Figure 7, then answer the questions that follow.



Figure 7: *Recreation and playground area*

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(a) In which direction is the treeplantation from the picnic area?

(b) In which direction is the football pitch from the picnic area?

(c) In which direction is the restaurant from the picnic area?

(d) From the football pitch, which direction will you take to reach the swimming pool?

Vocabulary

Analog	Anything that is not digital in its design and functioning.
Compass	A clock-like instrument that shows directions and has a arrow that always points North.
Digital	Anything that uses or works with computers or electronic devices to perform tasks.
Sunrise	The time of a day when the sun first appears in the morning.
Sunset	The time in the evening when the sun disappears below the horizon and night begins.

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Chapter Three

Drawing a simple map

Introduction

A map is one of the ways to present information about various features on the Earth's surface. In this chapter, you will learn the concept of map drawing, features that can be shown on a map, and map symbols. Also, you will learn map scale, map drawing tools, steps of drawing simple maps and digital maps. Similarly, you will learn to draw a map of your classroom, school, ward, district, and Tanzania. The competencies developed will enable you to draw a simple map of your surroundings.



Think

Features of the earth's surface that can be represented on a map.

The concept of map drawing

Map drawing is the art and science of representing the locations of features on the earth's surface on paper or any flat surface. A study of a map drawing is called cartography, and a person who draws maps is called a cartographer. In map drawing, symbols are used to represent features such as roads, houses, rivers, and mountains. This process emphasizes maintaining the correct proportions between the area, real objects, and the symbols placed on the map.

Features that can be shown on a map

Maps can display various features depending on their purpose. For example, a map can show roads that lead to different places, buildings

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like houses and schools, and rivers and oceans to represent water bodies. Also, maps can show mountains and valleys to represent higher and lower land, green areas like national parks and forests, city or country boundaries, and sports fields.

However, not everything on the earth's surface will be shown on a map. The features included on a map are those which are important to the map users. The cartographer decides what to include on the map based on the needs of the map users. Therefore, maps show only the key features necessary to make them easy to read and understand.



Activity 1

Visit the areas around your school, then list the important features that you would include if you were to draw a map of the school.

Map symbols

A map cannot show things exactly as they are in reality; instead, it uses map symbols. Map symbols are small drawings used on a map to represent real objects. When you want to show real objects on a map, you should use symbols to represent those objects. These symbols help to describe those features simply and understandably way. All symbols used on a map are shown in the map key, which explains all the symbols used. Figure 1 shows some of the symbols used in maps.

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River	Mountain
Tree	School
Railway	Mosque
Bridge	Church
Building	Hospital

Figure 1: *Map symbols*

Exercise 1

1. Observe the map of Mbuga Primary School, then identify the features represented by the symbols shown in the map key.

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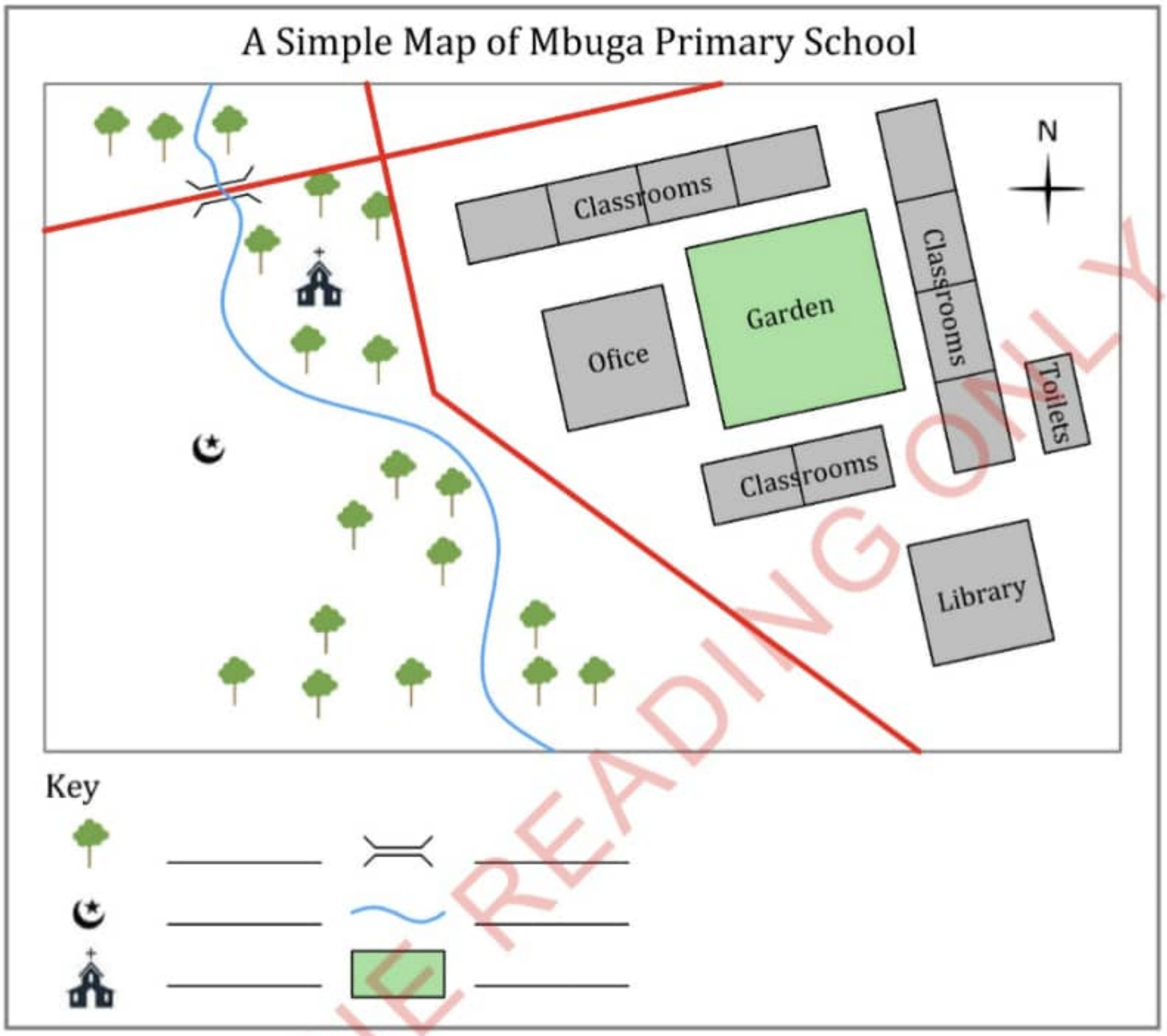


Figure 2: *Map of Mbuga Primary School*

2. Explain the importance of map symbols.
3. What features are not shown in a key? Draw those symbols in your exercise book.

Activity 2

Visit a library, take an atlas, then choose any map you like. Identify various map symbols and draw them in your exercise book.

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Map scale

A map scale is the ratio between the distance on the map and the real distance on the ground. The scale helps to show how the measurements on the map correspond to real-world measurements. The scale can be presented as a statement, representative fraction or ratio and linear scale.

As a statement

A statement scale describes the ratio of distance on the map to the real distance on the ground using words. For example, one centimetre on the map represents one kilometre on the ground.

As a Representative Fraction (R.F) or ratio

A representative fraction or ratio scale uses numbers to show the relationship between distance on a map and real distance on the ground, for example,

1:50000 or $\frac{1}{50000}$.

As a linear

A linear scale is a graphical representation of ground distance using a line. Figure 3 shows the linear scale.

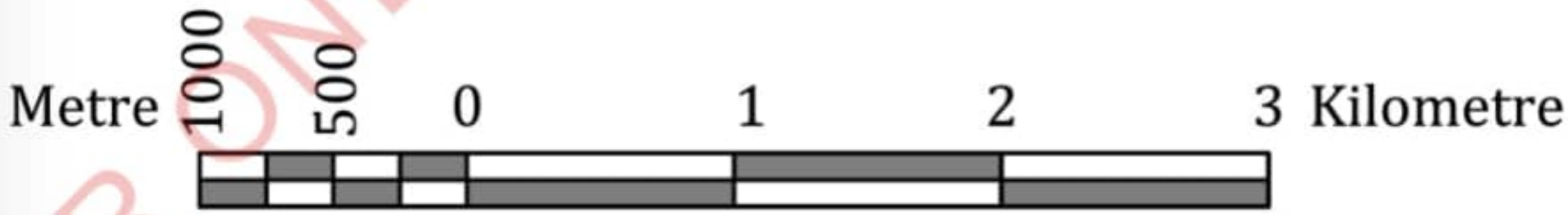


Figure 3: *Linear scale*

Map drawing tools

There are several tools used in drawing maps. These tools include a drawing table, paper, a pencil, a pair of dividers, a protractor, an eraser, a pen, a sharpener, colouring materials, and a ruler as shown in Figure 4.

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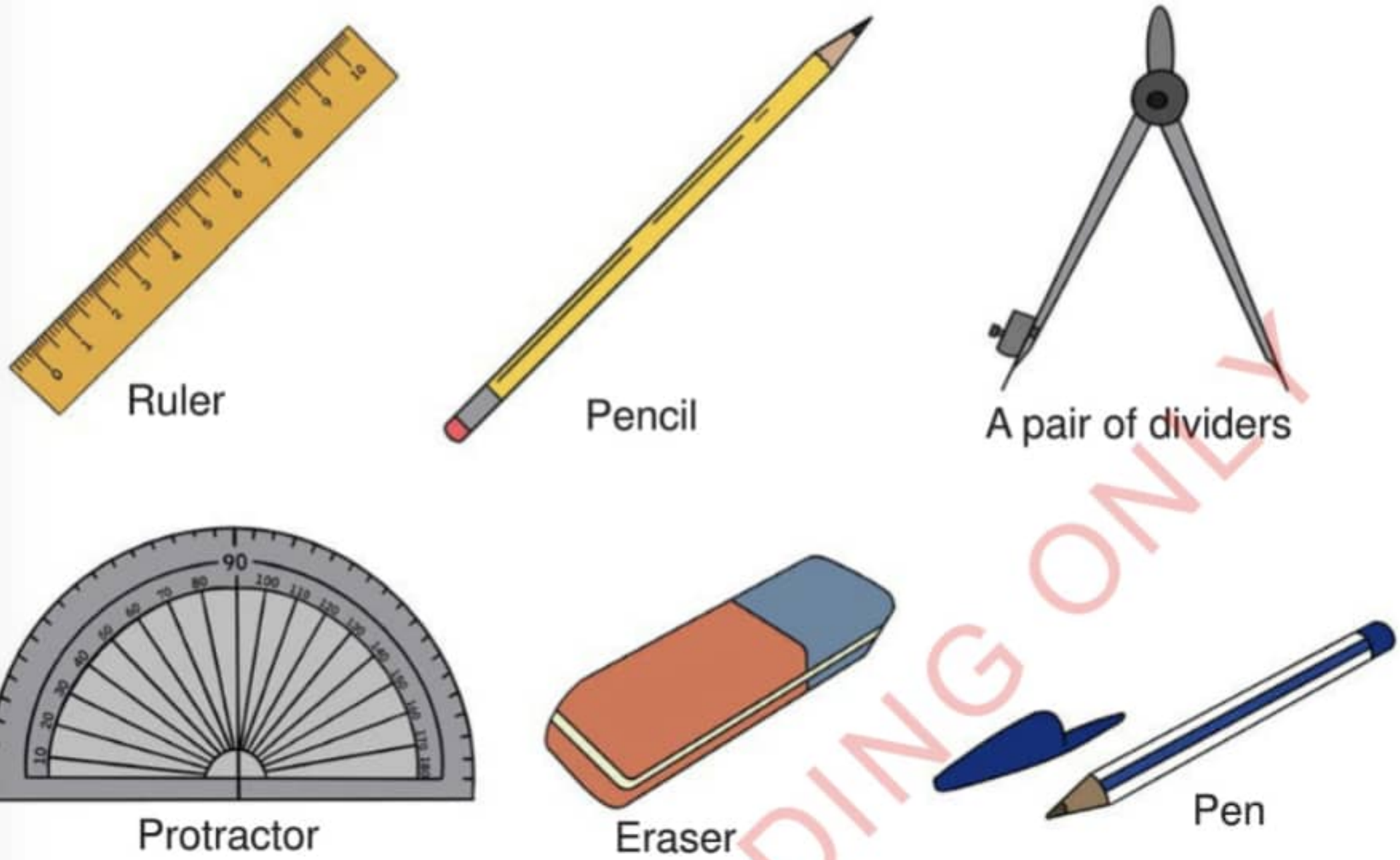


Figure 4: *Some of the map drawing tools*

Other tools that can be used in map drawing include a laptop, smartphone, tablet, digital drawing board, and digital drawing pen. Additionally, the cartographer may need drawing software such as Computer Aided Design (CAD) and Geographical Information System (GIS).

Drawing a simple map

A simple map is drawn without using a scale to show the location of features on the earth's surface.

Steps for drawing a simple map

(a) To draw a simple map, first identify the size, shape, and direction of the area to be shown on the map;

(b) identify the shapes, sizes, and locations of important features in the area you want to represent on the map;

(c) prepare the area where you will draw the map by sketching a square or rectangle on the paper. Ensure that you leave space for the map title, key, and other explanations;

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(d) place a North direction symbol in the upper right corner of the map to indicate the directions of North, South, East, and West;

(e) draw a shape within the square or rectangle you created earlier, representing the boundaries of the area to be mapped, in the correct direction;

(f) use symbols to represent the key features in the mapped area, ensuring they are oriented correctly. Start with the larger features and then add the smaller ones;

(g) prepare a key that explains all symbols used in the simple map;

(h) Write the title of the simple map;

(i) ensure that all important features are shown on the map and make adjustments as needed for improvement; and

(j) draw a square or rectangle (frame) around the map and all the important elements.

Drawing a simple map of a classroom

The following are steps to follow when drawing a classroom map showing objects such as the teacher's table and chair, student's desks or table and chairs, the door, windows, and other important items:

(a) Identify the size, shape, and direction of the classroom;

(b) Determine the shapes, sizes, and locations of important items in the classroom;

(c) Prepare the area where you will draw a map of the classroom by drawing a square or rectangle on the paper. Make sure you leave space for the map title, key, and other explanations; and

(d) Follow the steps of drawing a simple map to create the classroom map.

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Activity 3

Follow the steps for drawing a simple map, to draw a map of your classroom by showing all the important objects in the classroom.

Drawing a simple map of a school

A school map is a drawing that shows the layout of the school, including playgrounds, buildings such as classrooms, and teacher's offices, and other important areas. The following steps will help you to draw a school map:

(a) Identify the size, shape, and direction of the school;

(b) determine the size, shapes, and locations of important features within the school area;

(c) prepare the area where you will draw the school map by drawing a square or rectangle on the paper. Make sure to leave space for the map title, key, and other important explanations; and

(d) follow the steps of drawing a simple map to draw the school map.

Activity 4

Draw a simple map showing your school environment by following all the steps for map drawing.

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Exercise 2

1. What essential tools should you prepare when you are drawing a map?
2. Draw the symbols used to represent the following features:
   (a) Buildings;
   (b) trees;
   (c) road;
   (d) mosque;
   (e) church; and
   (f) bridge.
3. Explain the difference between a classroom map and a school map.
4. Explain the function of a key on a map.

Drawing a map of a ward

A ward is an administrative area that includes several villages or streets. Drawing a ward map involves showing the boundaries of the ward along with various geographical features within that specific area. Examples of such features include hospitals, schools, churches, mosques, roads, and rivers located within the ward.

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Activity 5

Observe Figure 5, then answer the questions that follow.

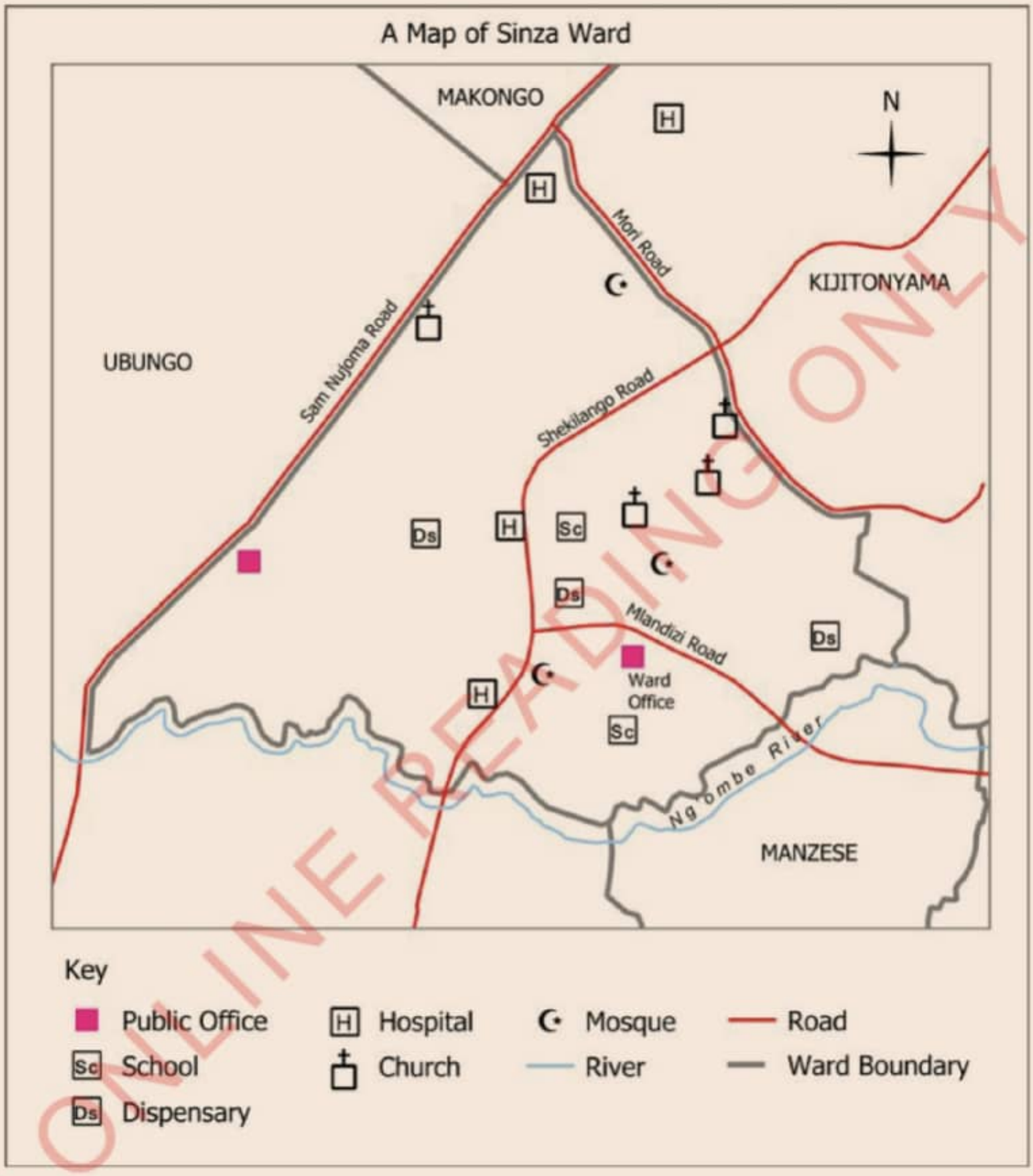


Figure 5: *A map of Sinza Ward*

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Questions

1. Identify the features represented on a map of Sinza Ward.
2. Identify the key elements of the map that have been considered in that map.
3. Draw a map of Sinza Ward, then show the following:
   (a) A road that is in the Southwest of the map;
   (b) A mosque that is in the South of the map;
   (c) A ward that is in the North of the map;
   (d) A river that is in the South of the map;
   (e) A hospital that is in the Southwest of Kijitonyama; and
   (f) An office that is in the South of Mlandizi Road.

Drawing a map of a district

A district is an administrative division of a certain area within a country. A district comprises of several wards. Each district is led by a District Commissioner who is appointed by the President.

Activity 6

Observe Figure 6 then; answer the questions that follow:

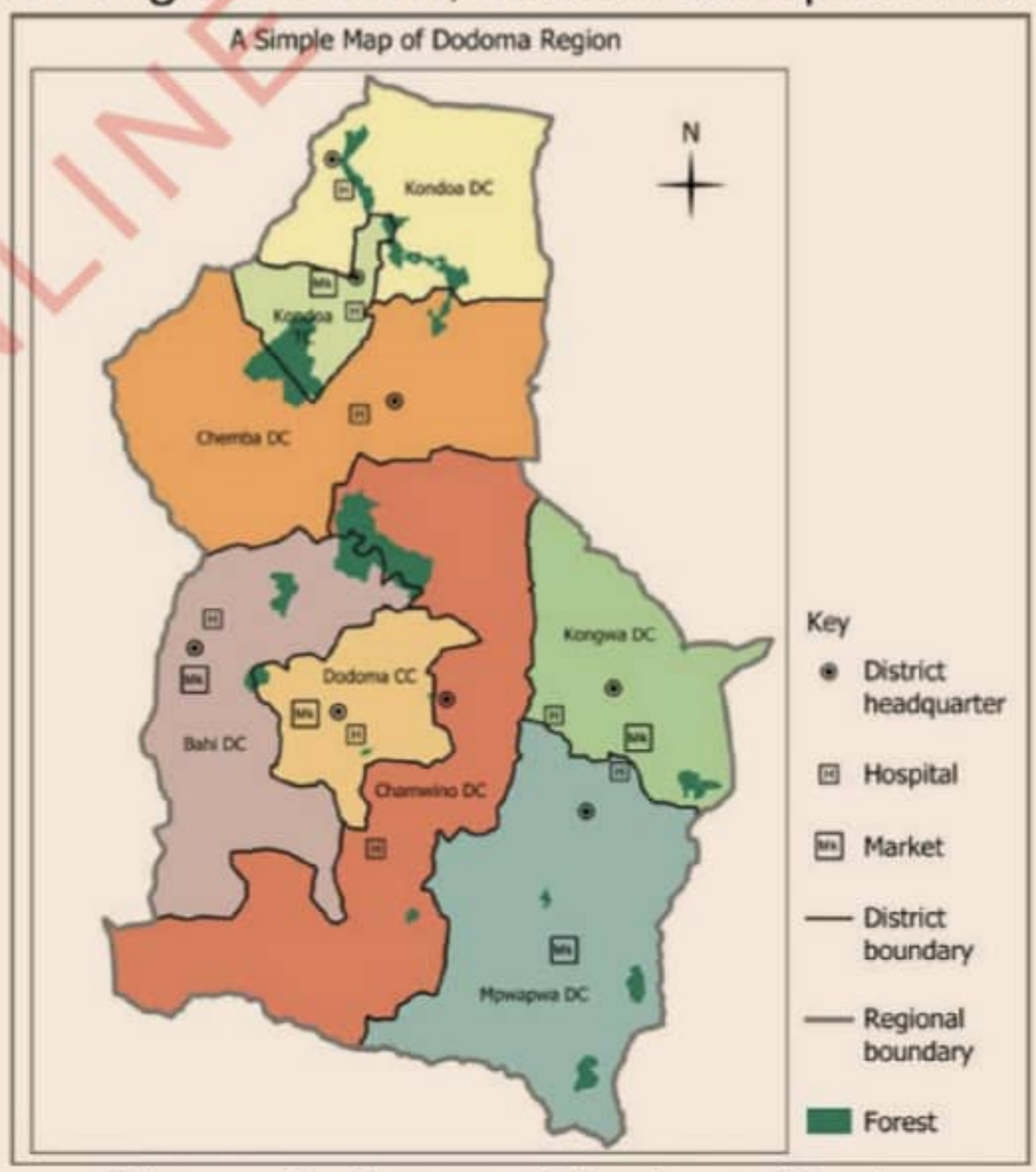


Figure 6: *A map of Dodoma Region*

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Questions

1. Choose one of the districts, then draw it in your exercise book.
2. On the district map you have drawn, indicate the following:
   (a) The district headquarters;
   (b) A forest;
   (c) A market; and
   (d) The district hospital.

Activity 7

Use an atlas or any reliable online sources to find a map of Tanzania, that shows administrative boundaries. Then, identify the region where you live and do the following:

(a) Draw a map of the region where you live;
(b) Show the districts of that region;
(c) Shade the district where you live;
(d) Place a North arrow to show direction; and
(e) Add a map key.

Drawing a map of Tanzania

The map of Tanzania provides various types of information about the country. The information shown on a map of Tanzania varies depending on its purpose. For example, a map of Tanzania might show the location of the capital city, Mount Kilimanjaro, railways, the ocean and lakes, regional boundaries, and borders with neighbouring countries.

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Activity 8

1. Use an atlas or any reliable online sources to find a map of Tanzania, then draw it and show the following features:
   (a) Mount Kilimanjaro;
   (b) The Indian Ocean;
   (c) Lake Victoria;
   (d) Lake Tanganyika;
   (e) Lake Nyasa; and
   (f) The Islands of Pemba, Unguja, and Mafia.
2. Add a key, a North direction, a frame, and a title to the map you have drawn.

Exercise 3

1. What are the benefits of knowing how to draw a map of Tanzania?
2. In the map of Tanzania, what important features are found on the borders of Tanzania and neighbouring countries?

Digital map Drawing

Digital map drawing involves using computer technology or smartphones with map drawing softwares like ArcGIS, QGIS, AutoCAD, and Google My Maps. Each software has unique features for map drawing.

At this level, you will learn to draw maps digitally using the Google My Maps software. Drawing maps with Google My Maps software is easier, whether you use the website or the smartphone app. Google My Maps has an interface that allows users to interact with the tool easily and draw maps quickly without needing much expertise.

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Things to consider

The following are some of the things to consider when drawing a digital map using "Google My Map" software:

1. Ensure you have access to a device such as a smartphone, tablet, or computer;
2. make sure the device is connected to the internet;
3. identify the geographical area you want to map, for example, country boundaries, roads, or lakes; and
4. make sure that you have a Google account by registering first. Use the "Google My Maps" software to register. This will allow you to create maps, including drawing shapes and adding important symbols of various features.

Steps for drawing maps using Google My Maps' software

The following steps will guide you in drawing a map using Google My Maps software. The map of Swaga Swaga Game Reserve in Dodoma region is used as an example.

1. Open your browser and go to Google. Type "Google My Maps" into the search bar, as shown in Figure 7. Then, click on "My Maps" to open Google My Maps

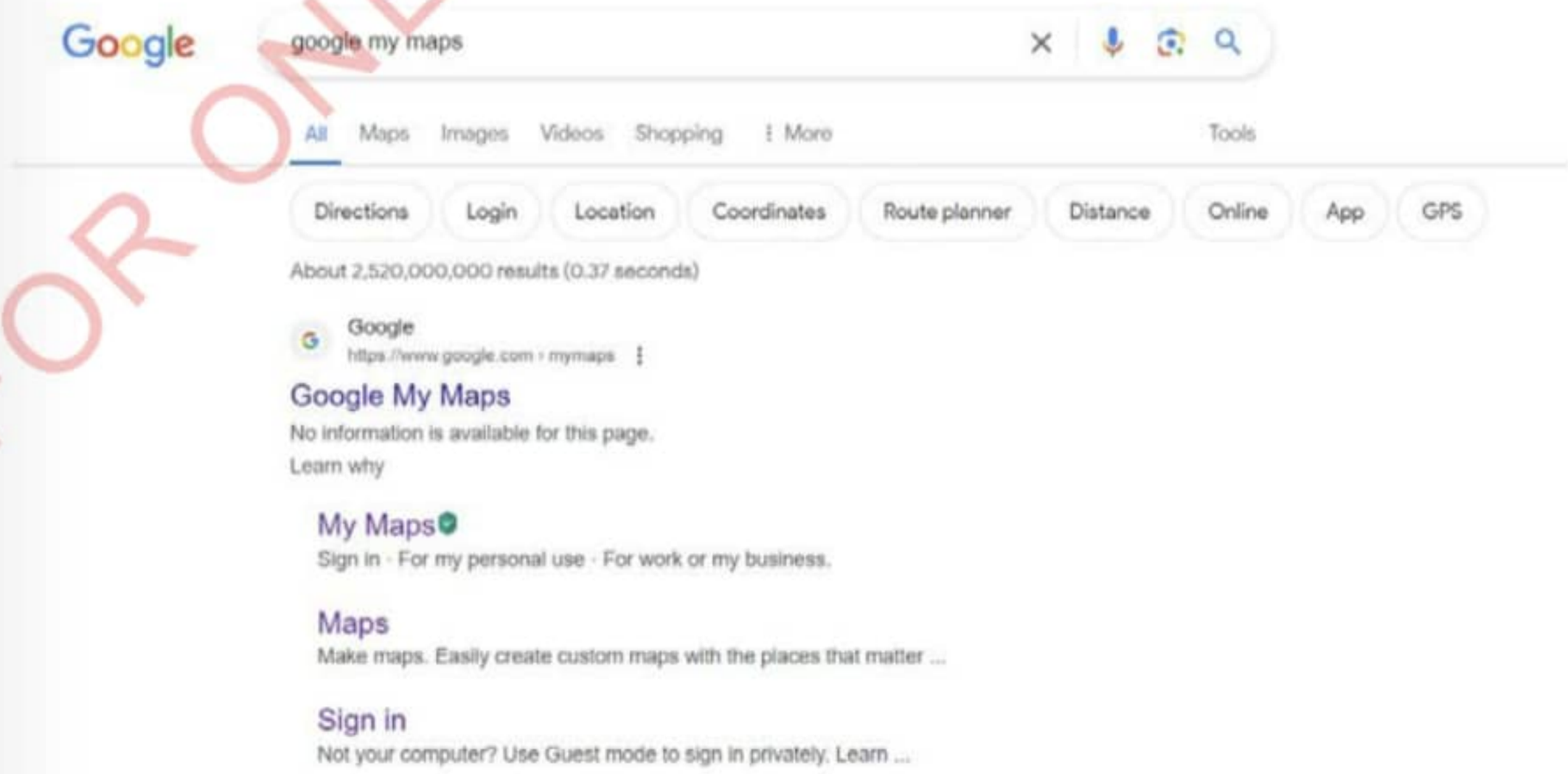


Figure 7: *Website browser*

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2. Go to the ‘Create a New Map’ menu in ‘Google My Maps’, as shown in Figure 8



Figure 8: *Menu for creating maps in Google My Maps*

3. Search for the name of the location you need on the map, such as parks, reserves, villages, wards, districts, or regions. For example, type “Swaga Swaga Game Reserve,” as shown in Figure 9.



Figure 9: *Google My Maps interface for searching geographical area or location*

4. Click the search button to display Swaga Swaga Game Reserve, as shown in Figure 10. Once Swaga Swaga Game Reserve is displayed, you can draw its boundary. To start drawing the boundary, click “Draw line” (1) and then “Add line or shapes” (2), as shown in Figure 10.

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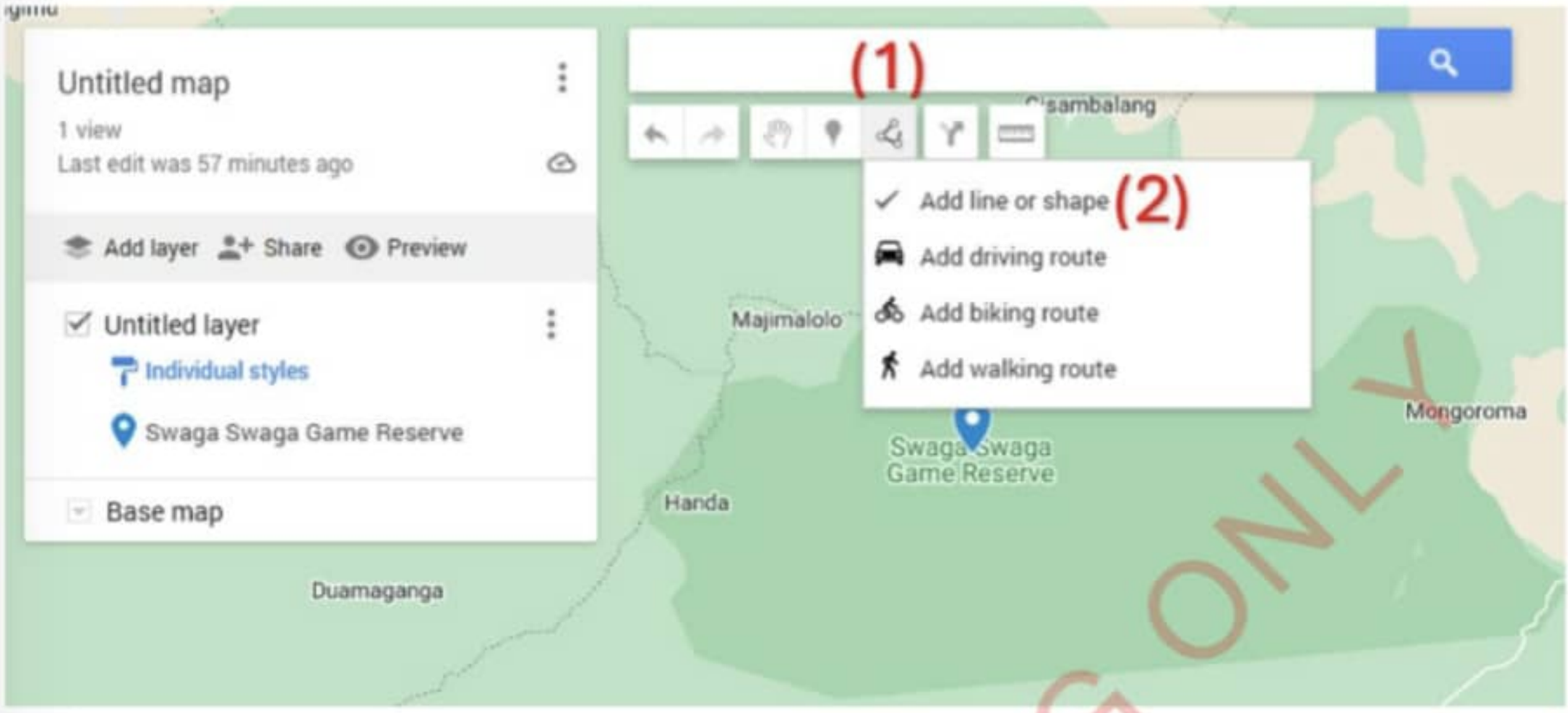


Figure 10: *Selecting different drawing tools*

5. Start drawing the map by tracing the boundary of Swaga Swaga Game Reserve, as shown in Figure 11.



Figure 11: *Drawing a boundary of Swaga Swaga Game Reserve*

6. Continue drawing until you return to the starting point to complete the boundary of Swaga Swaga Game Reserve. Once the boundary is complete, the cursor will change from an additional sign to a pointing finger. Click the left mouse button or double-tap the right mouse button if using a touch screen to complete the drawing. Finally, type the word “Swaga Swaga” in place of “Polygon 1” and click “Save,” as shown in Figure 12.

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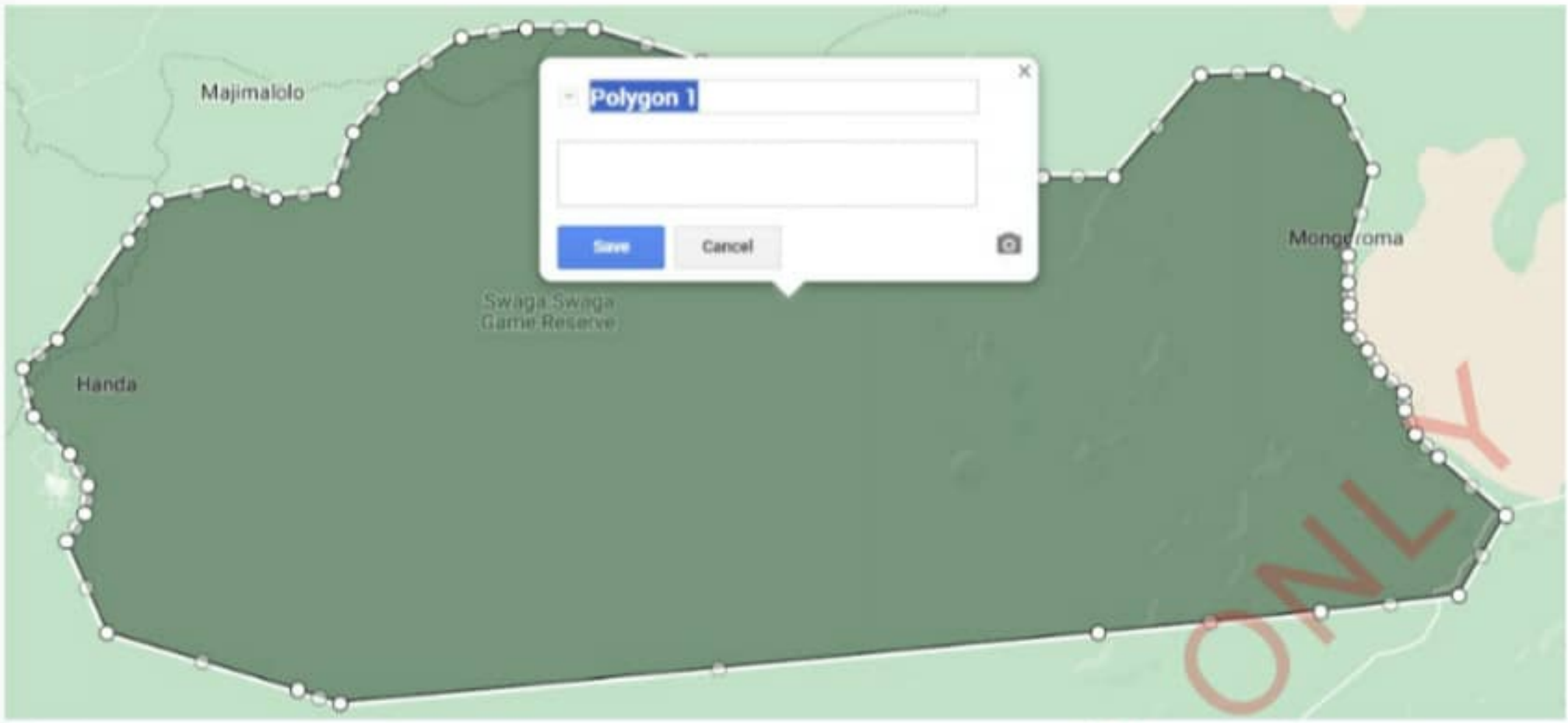


Figure 12: *A Complete outline of the map of Swaga Swaga Game Reserve*

7. You can change the title to match the content of the map. For example, use “Map of Swaga Swaga Game Reserve.” Go to the window on the left side of the map, select “Untitled Map,” then write “Swaga Swaga Map” in the box labelled “Map Title” and click the “Save” button.

8. Change the appearance of your map by using different available colours. Click the left mouse button inside Swaga Swaga Game Reserve, then click the “Style” button and choose your favourite colour, for example the green colour, as shown in Figure 13.

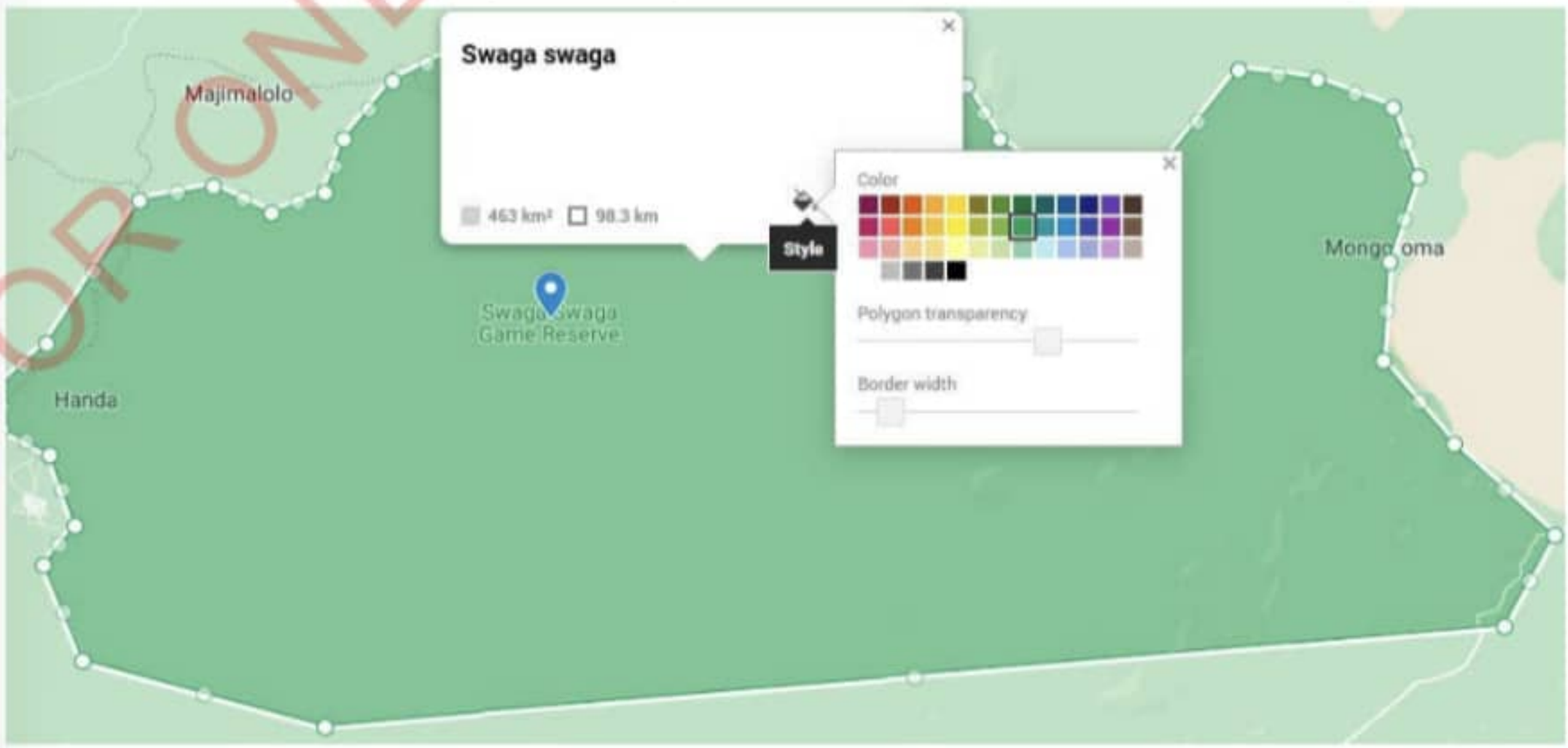


Figure 13: *Changing the colour on the map*

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9. To add a map key, left-click the three vertical dots at the bottom (1) on the left side of the map. Select “Rename this layer” (2), type “Key” in the “Edit layer name” box (3), and then click the “Save” button (4), as shown in Figure 14.

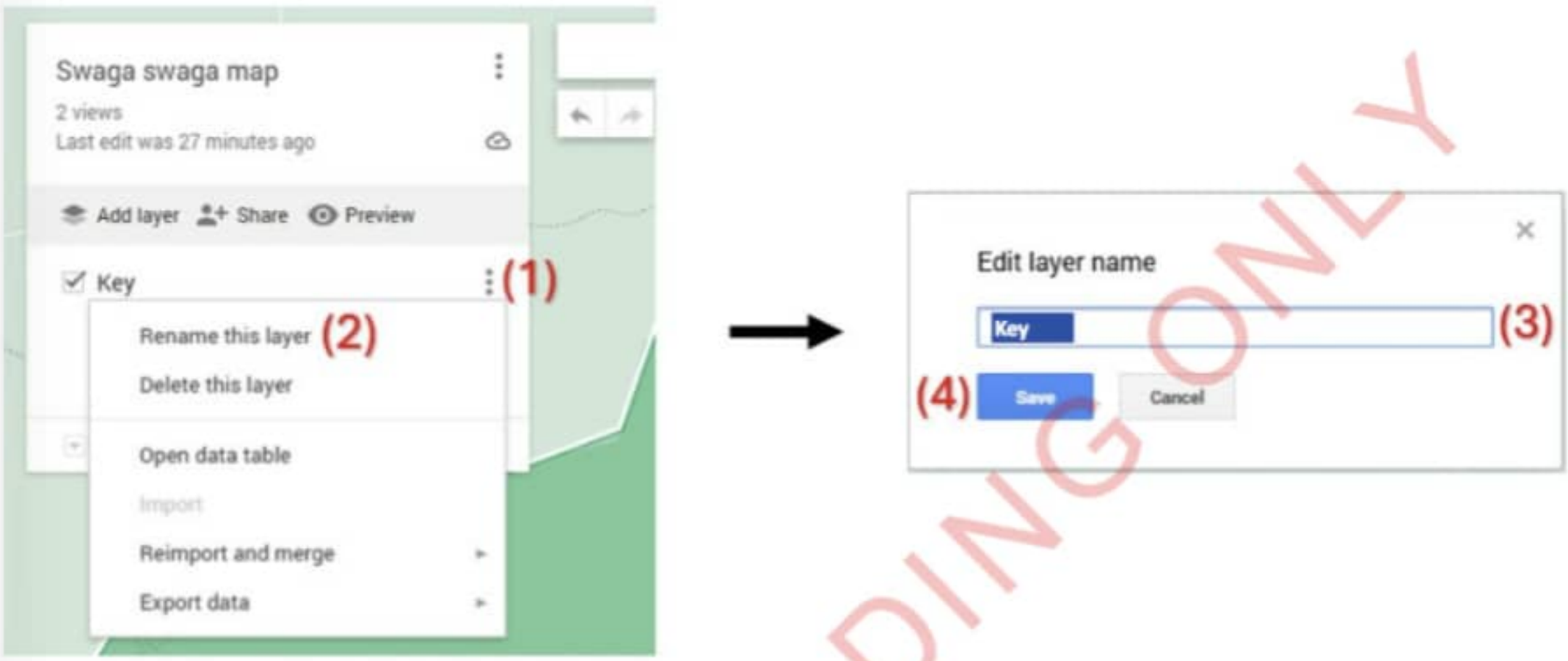


Figure 14: *Inserting a key on the map you have drawn*

10. Print your map for future use. You can print a digital map in two ways: as a PDF file or as an image. To do this, go to the three dots (1) on your map, then select “Print map” (2), as shown in Figure 15.

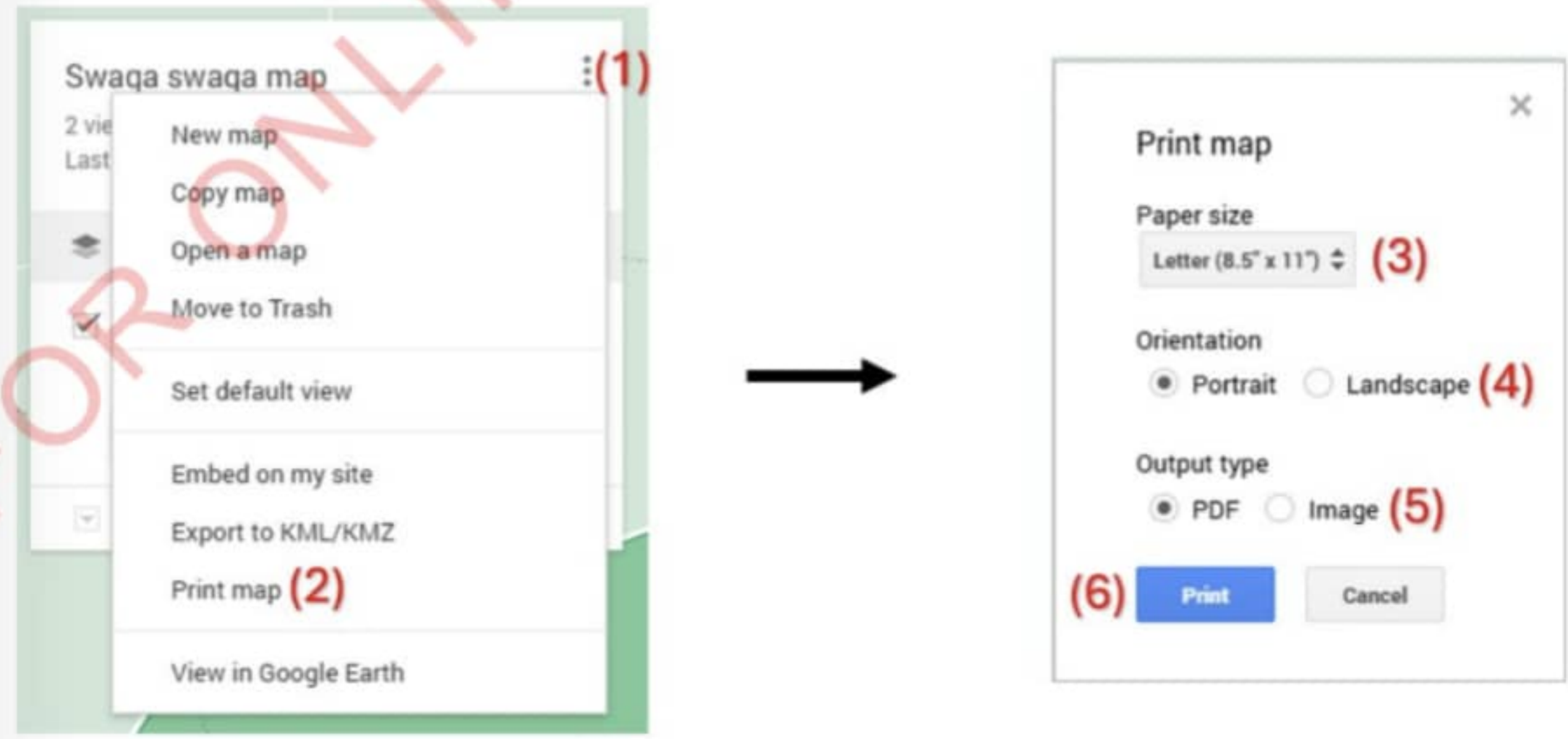


Figure 15: *Printing a map*

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11. Select the paper size (3), paper orientation (4), and format for printing (either PDF or image) (5). Then click “Print” (6) to save the map as a PDF or image, as shown in Figures 15 and 16.

Swaga swaga map



Figure 16: *A map of Swaga Swaga Game Reserve*

Activity 9

1. Choose an area within your region or a famous national park in Tanzania, and draw it digitally.
2. Explain the things you considered while drawing your map.
3. Which essential elements of a map are missing in your digital map?

Exercise 4

1. What softwares are used for digital map drawing?
2. Name the essential requirements for digital map drawing.
3. How can you print a digital map you have drawn?

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Revision exercise

Choose the letter of the correct answer.

1. The scale of 1:100000 can be expressed in words as follows:
   (a) One centimetre on the map represents one hundred kilometres on the ground.
   (b) One kilometre on the map represents one hundred thousand centimetres on the ground.
   (c) One centimetre on the map represents one hundred thousand centimetres on the ground.
   (d) One hundred thousand centimetres on the map represents one centimetre on the ground.

2. Which software was used to draw the Swaga Swaga map in chapter three?
   (a) ArcGIS.
   (b) QGIS.
   (c) AutoCAD.
   (d) Google My Maps.

3. Which drawing tool do you use when tracing a road on Google My Maps?
   (a) 'Add direction'.
   (b) 'Add line or shape'.
   (c) 'Add marker'.
   (d) 'Draw polygon'.

4. The study of map drawing is known as _____

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(a) map drawing.

(b) map design.

(c) cartography.

(d) cartographer.

5. Which of the following colours represent vegetation?

   (a) Black.

   (b) Green.

   (c) Dark colour.

   (d) Brown.

6. Explain how a map drawn on a paper can represent a large area of land.

7. Explain the steps to be followed when drawing a simple map.

8. Draw a digital map of your ward.

9. Draw map symbols representing the following:

   (a) Railways;

   (b) Footpath;

   (c) School; and

   (d) Mountain.

10. Draw a map of your home area, showing the objects located to the North, South, East, and West of the house.

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Vocabulary

Browser	Is a software that lets you access and view websites on the internet
Interface	A device or program that enabling a user to communicate with a computer.
Mouse	Is a handheld input device used to control a computer by moving a pointer on the screen and clicking on items.

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Chapter Four

The uses of map

Introduction

In life, sometimes we need to travel from one place to another that we are not familiar with, to access services, relax, or visit relatives and friends. It can be difficult to reach the intended place if we do not have someone to guide us. However, a map is a useful tool that can help you to find the route and guide you to your destination. In this chapter, you will learn how to use a map to find directions, measure distance, identify administrative boundaries, and determine relief of the Earth's surface. You will also learn how to use maps to determine the population of a particular place. The competencies developed will enable you to use maps in different contexts.



Think

Reaching unfamiliar places using a map

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Uses of maps to determine direction and location

Activity 1

Observe Figure 1, then answer the following questions:

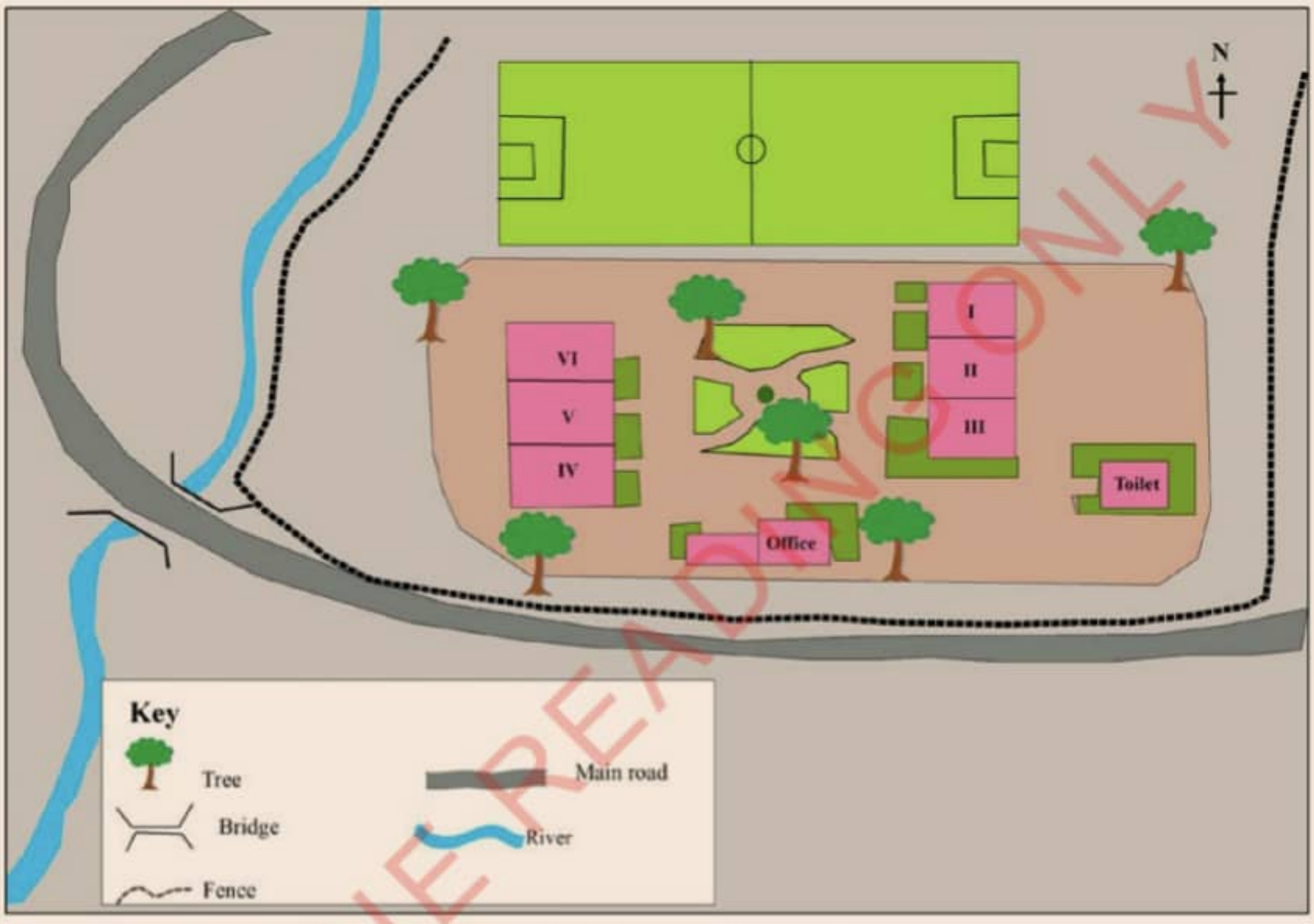


Figure 1: *Map of the school*

Questions

1. What things are located to the North of the main road?
2. In which direction is the football ground located from classrooms?
3. On which direction is Class V located from Class VI?
4. On which direction is Class IV located from Class II?
5. Which direction of the school area is not fenced?

The direction and location of features on a map can be identified using the symbol of the North direction. The North direction shows which side is North, South, East, or West. This will help you determine the location of different features on the earth's surface. It will also help you to find the directions from one place to another.

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Similarly, we can find directions and locations using digital maps and GPS. GPS helps you determine where you are, while a digital map shows you where you want to go. After determining your present location and identifying your destination, the GPS calculates and shows the distance and direction to follow. Digital maps and GPS are found on electronic devices such as smartphones, tablets, and computers. Different groups of people, including ship captains, drivers, pilots, geologists, and land surveyors, use maps in their work to determine locations, distances, and directions.

Activity 2

Observe Figure 2, then answer the following questions using the North direction:

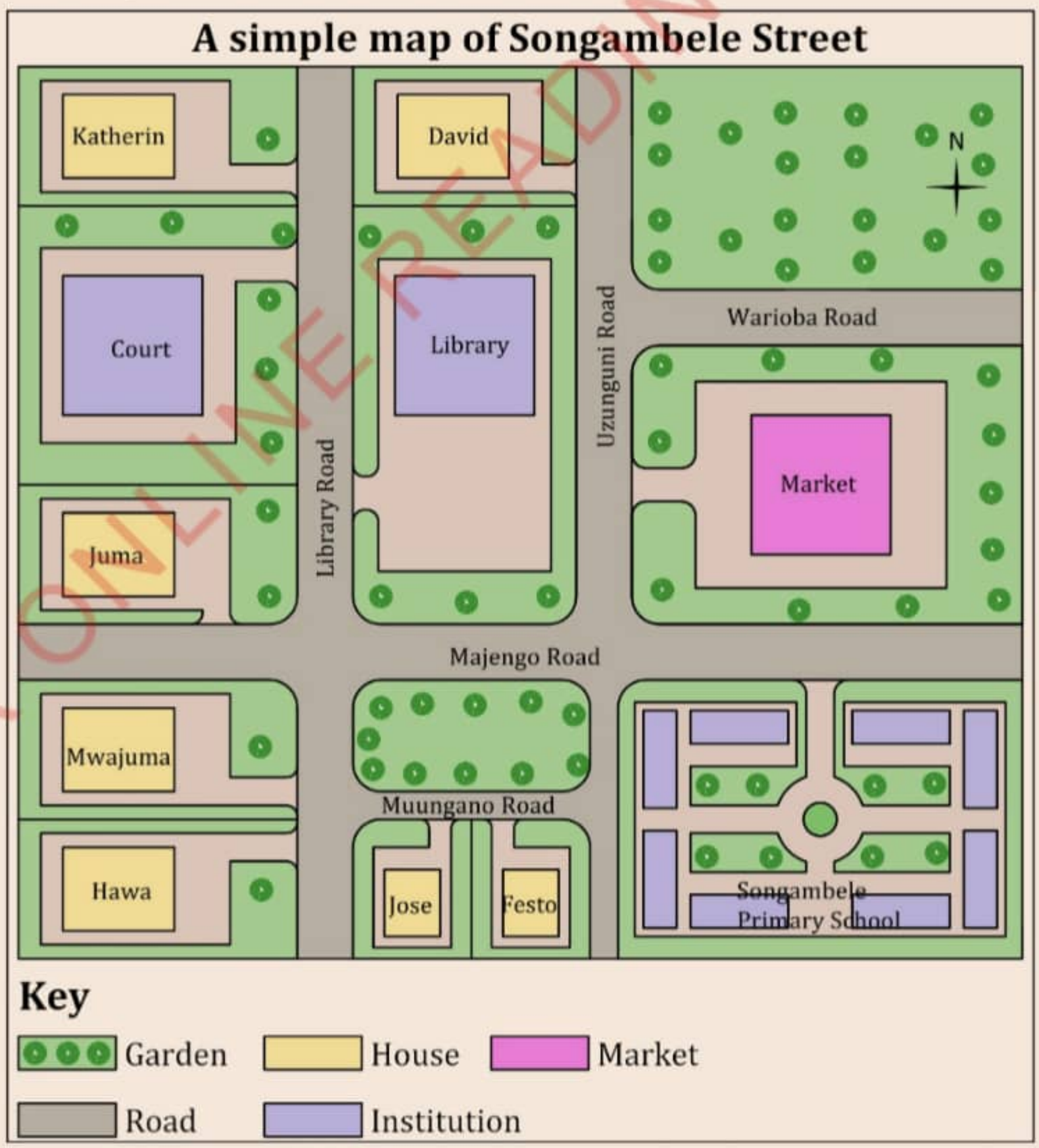


Figure 2: *Songambele street*

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Questions

1. In which direction is the school from Juma's house?
2. In which direction is David's house from Mwajuma's house?
3. Which direction will the driver Masanja take when driving from Warioba Road to the court?
4. How would you direct your friend to Songambele Primary School from the garden North of Warioba Road?
5. In which direction is David's house from Katherin's house?

Uses of maps to determine distance

Activity 3

Observe Figure 3, then answer the following questions:

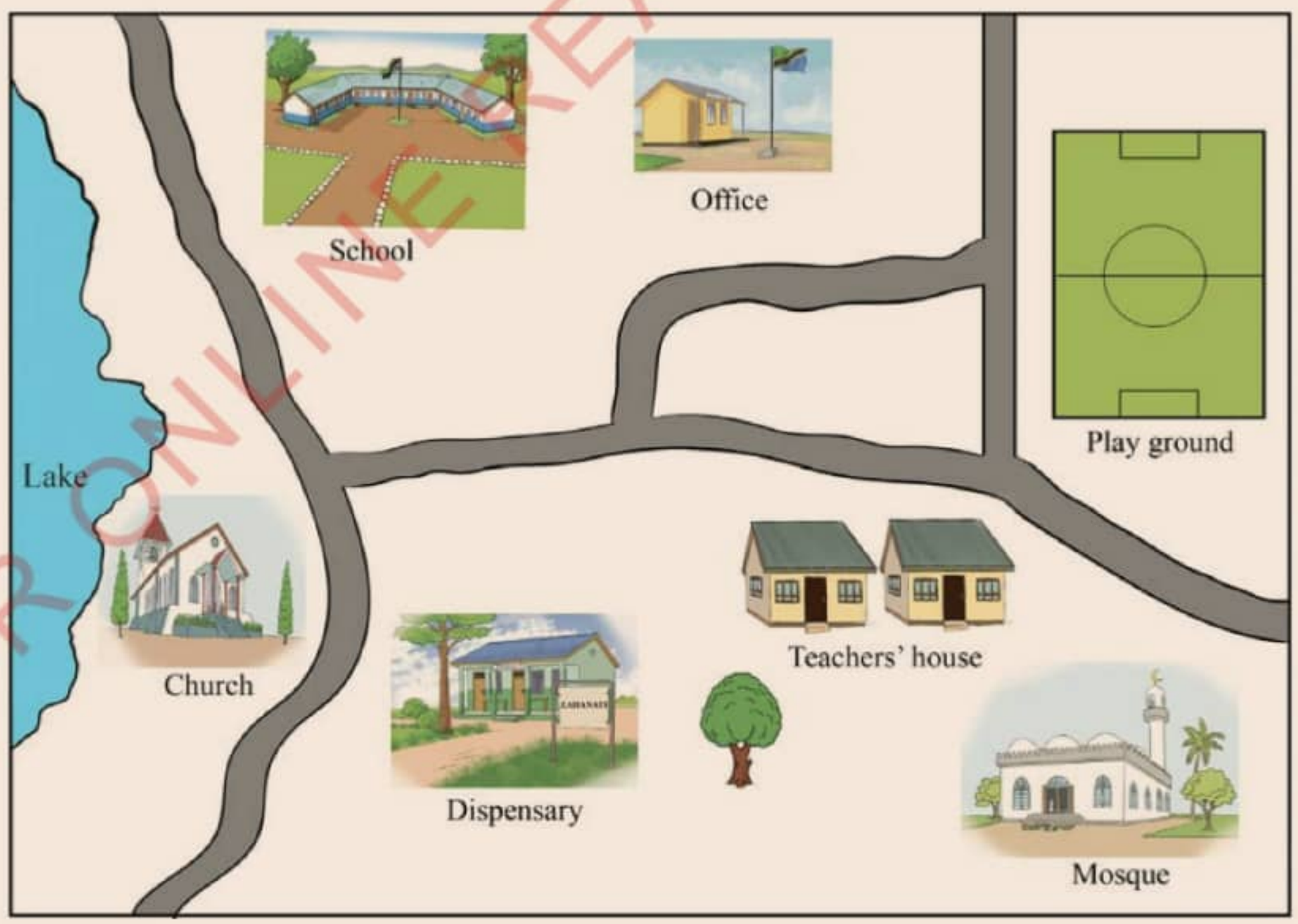


Figure 3: *Makuti Street*

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Questions

1. Which building is nearest to the lake?
2. Which building is nearest to the dispensary?
3. Which building is near the office?
4. What other buildings are near to the lake?
5. Identify the building that is farthest from the football ground.

You can determine the distance from one place to another on a map by following these steps:

(a) Measure the distance on the map between the two places; and

(b) Convert the map distance into the ground distance using the scale.

This measurement can include both straight and curved features and can be done with both digital and non-digital maps.

Measurement of straight distance

On a map, straight distance is measured using a piece of paper, thread or string, a ruler, a pair of dividers, and a scale. In this section, a piece of paper, a ruler, and a scale will be used.

1. *Using a piece of paper*

The following steps are involved in measuring distance on a map using a piece of paper and scale:

(a) Draw a straight line connecting two points on a map, for example, point A and B as shown in Figure 4.



Figure 4: *Straight-line connecting points A and B*

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(b) Take a piece of paper and fold it to create a straight edge. Place the edge parallel to the straight-line AB and accurately mark and label the points as A and B on the paper. Make sure it aligns perfectly with points A and B as shown in Figure 5.

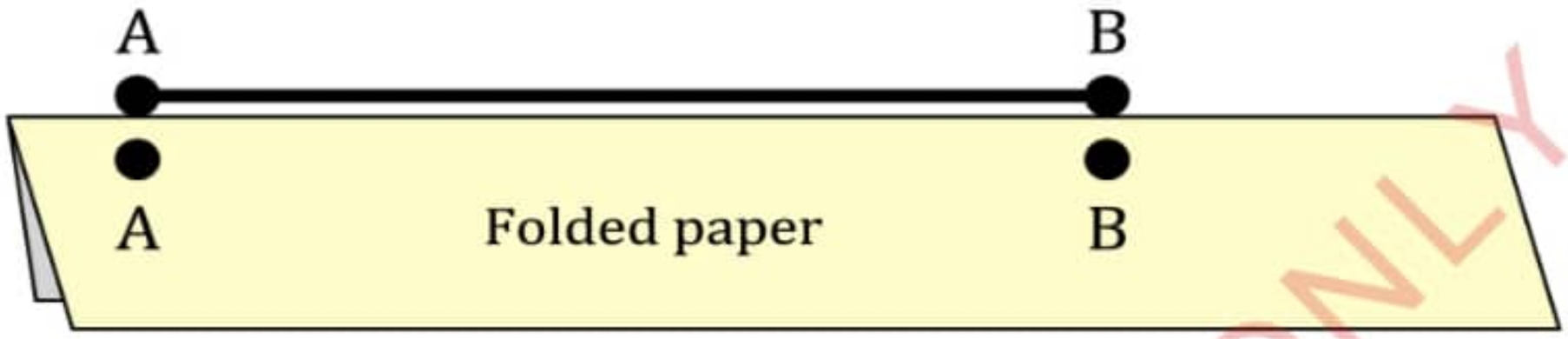


Figure 5: *Measurement of distance from points A and B using a piece of paper*

(c) A map distance from point A to B marked on the folded paper can be converted into real ground distance using a linear scale or ratio scale. To use a linear scale, place the folded paper along the scale by aligning point A with zero (0) and point B with the corresponding point on the right side of the linear scale. If point B is aligned with a certain number on the scale, as shown in Figure 6, that number represents the real ground distance between points A and B. In this case, the ground distance from point A to B is 3 kilometres.

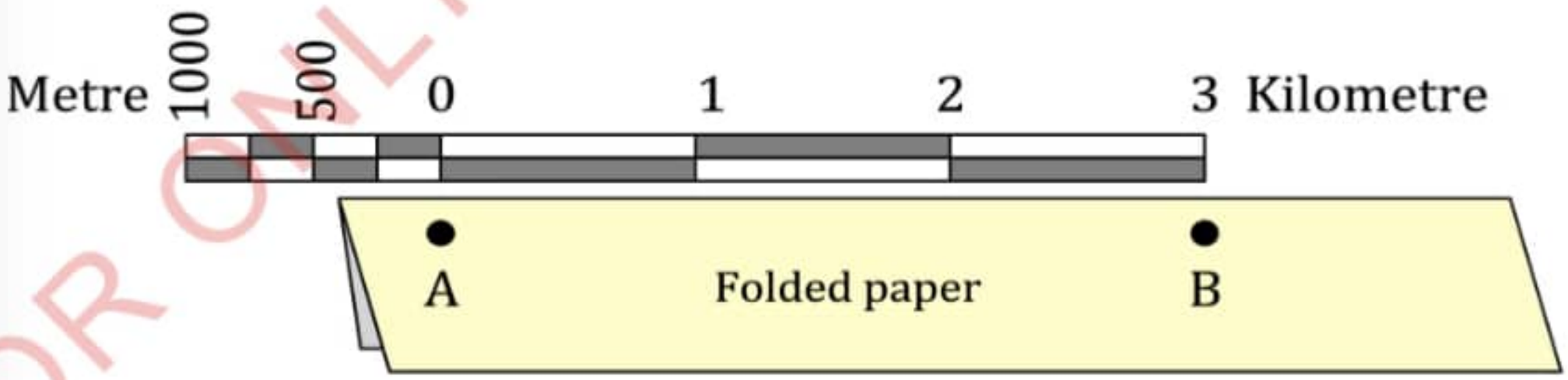


Figure 6: *Point B is aligned with the number 3, representing a distance of 3 kilometres from point A to point B*

If point B is between two numbers on the linear scale, mark and label the number on the left side of point B (1) on the folded paper, as shown by the red mark and number in Figure 7(a).

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Then, move the folded paper to the left along the linear scale by aligning point B with zero (the red mark will be on the left side of the linear scale). Observe the number in the linear scale that is nearest or aligned to the red mark and label it as shown in Figure 7(b). From Figure 7(a), the number on the left side of point B is 2 and the unit is km, so it is 2 km. From Figure 7(b) the nearest number to the red mark is 500 and the unit is m, so it is 500m. Therefore, the ground distance from point A to B is 2 kilometres and 500 metres.

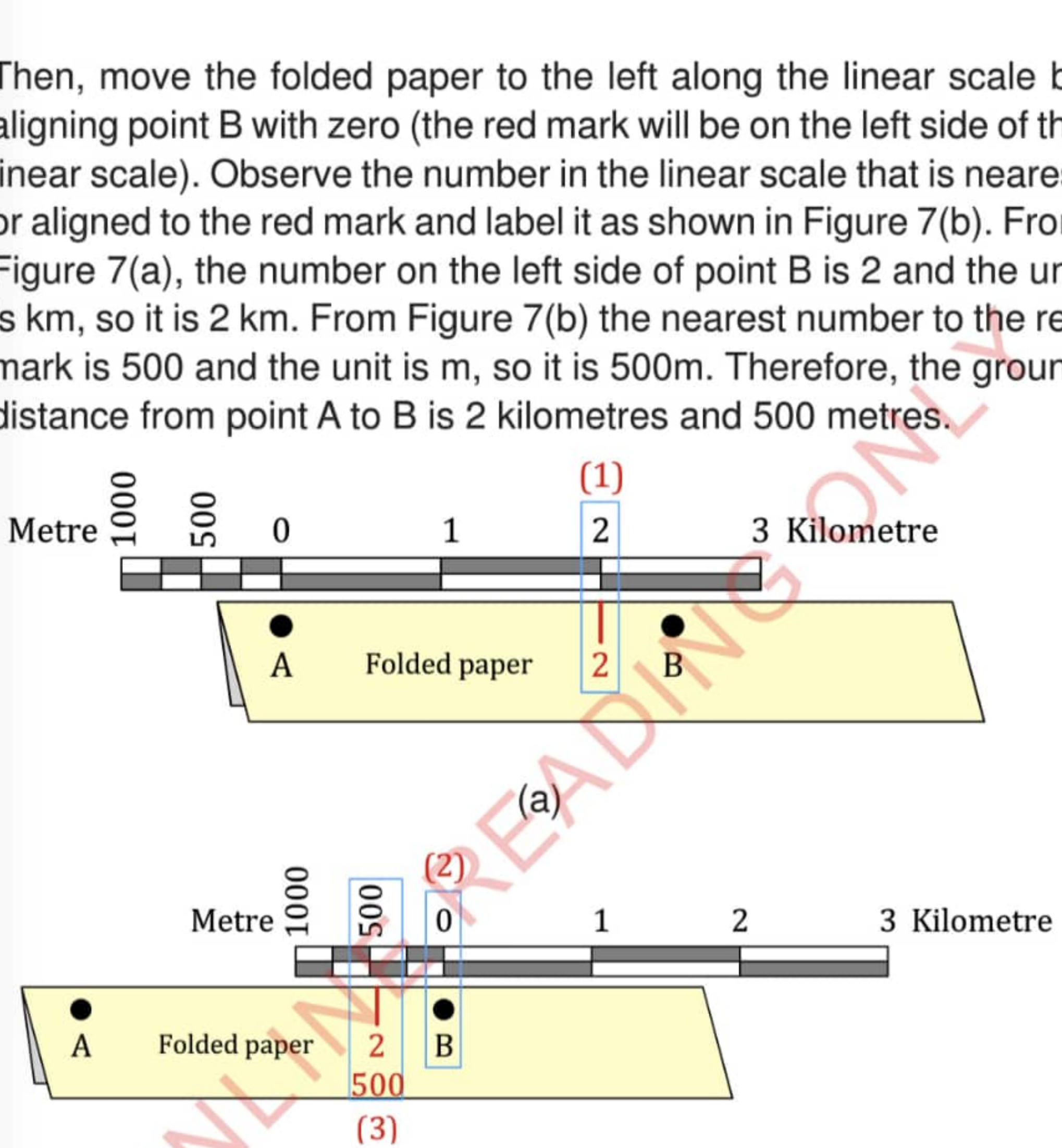


Figure 7: *Convert the map distance to ground distance when point B is between two numbers on the right side of the linear scale*

(d) Using a ratio scale, measure the map distance from points A to B on the folded paper with a ruler. For example, if, the map distance between points A and B is 5 centimetres and the ratio scale is 1:50000, the real distance on the ground from point A to point B can be calculated as follows:

distance on the ground (Km) =

Distance on a map (cm) x Scale denominator

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= 5 centimetres x 50000 centimetres

= 250000 centimetres

Since 1km = 100000 cm, the ground distance can be converted from centimetres to kilometres as follows:

Distance on the ground (Km) =

$$\left(\frac{\text{Distance on the ground in centimetres}}{\text{100000 centimetres}}\right) \times 1\text{Kilometres}$$

$$= \frac{\text{250000 centimetres} \times 1\text{kilometre}}{\text{100000 centimetres}}$$

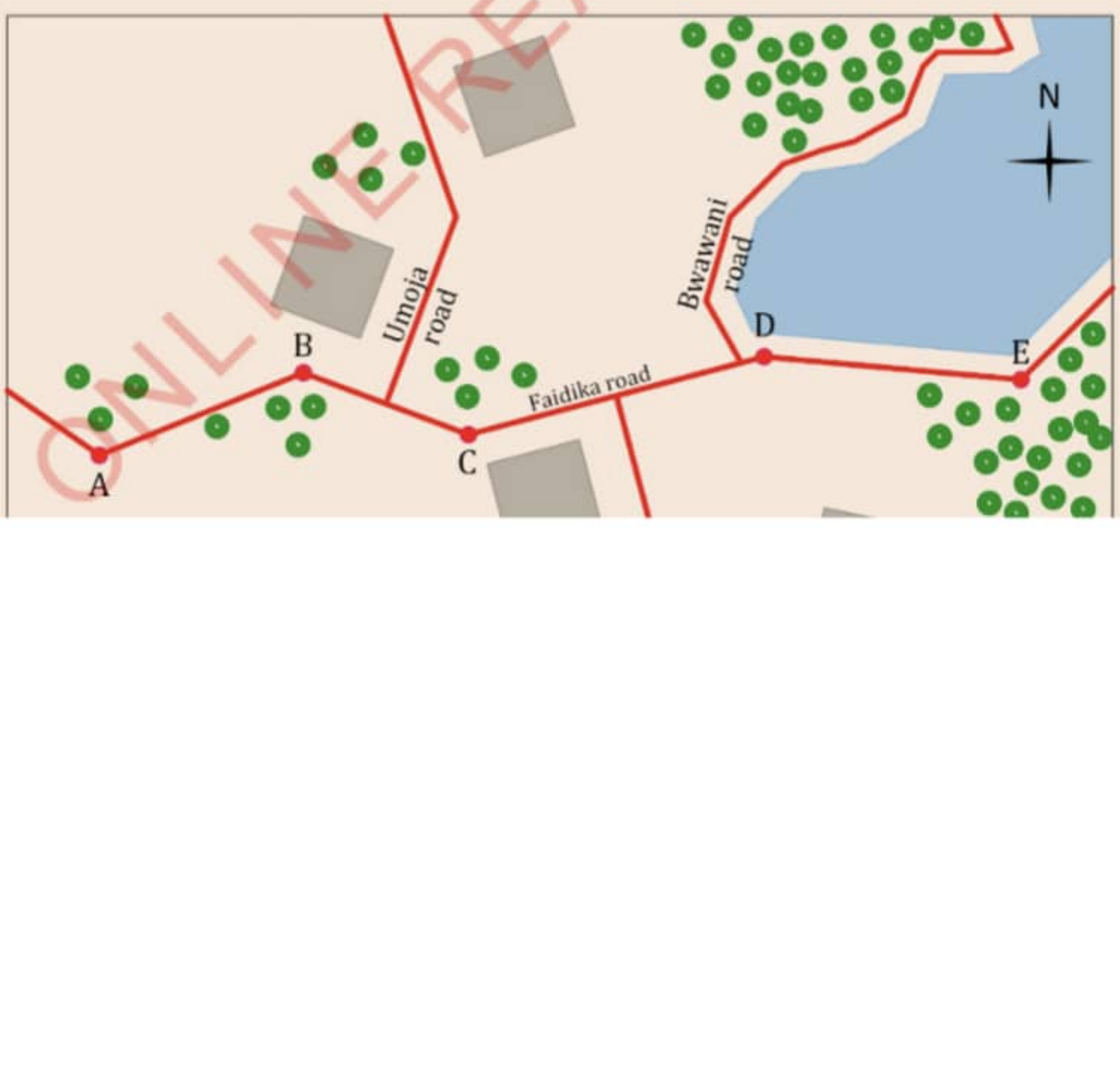
= 2.5 kilometres

Therefore, the distance on the ground is 2.5 kilometres.

Activity 4

Observe Figure 8, then answer the following questions

A simple map of Faidika



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Questions

By using a piece of paper;

(a) Calculate the distance from B to C.

(b) Calculate the distance from C to E.

(c) Calculate the distance from A to D.

(d) Calculate the distance from B to E.

(e) Calculate the distance of Faidika road from point A to E

2. Using a ruler

The following steps are involved in determining distance on a map using a ruler:

(a) Draw a straight line connecting two points, for example, points A and point B as shown in Figure 9.



Figure 9: *A straight line connecting point A and point B*

(b) Lay a ruler along a straight line AB, align the zero mark on a ruler with point A (1), and mark on ruler where point B is located as shown by red mark (2) in Figure 10.

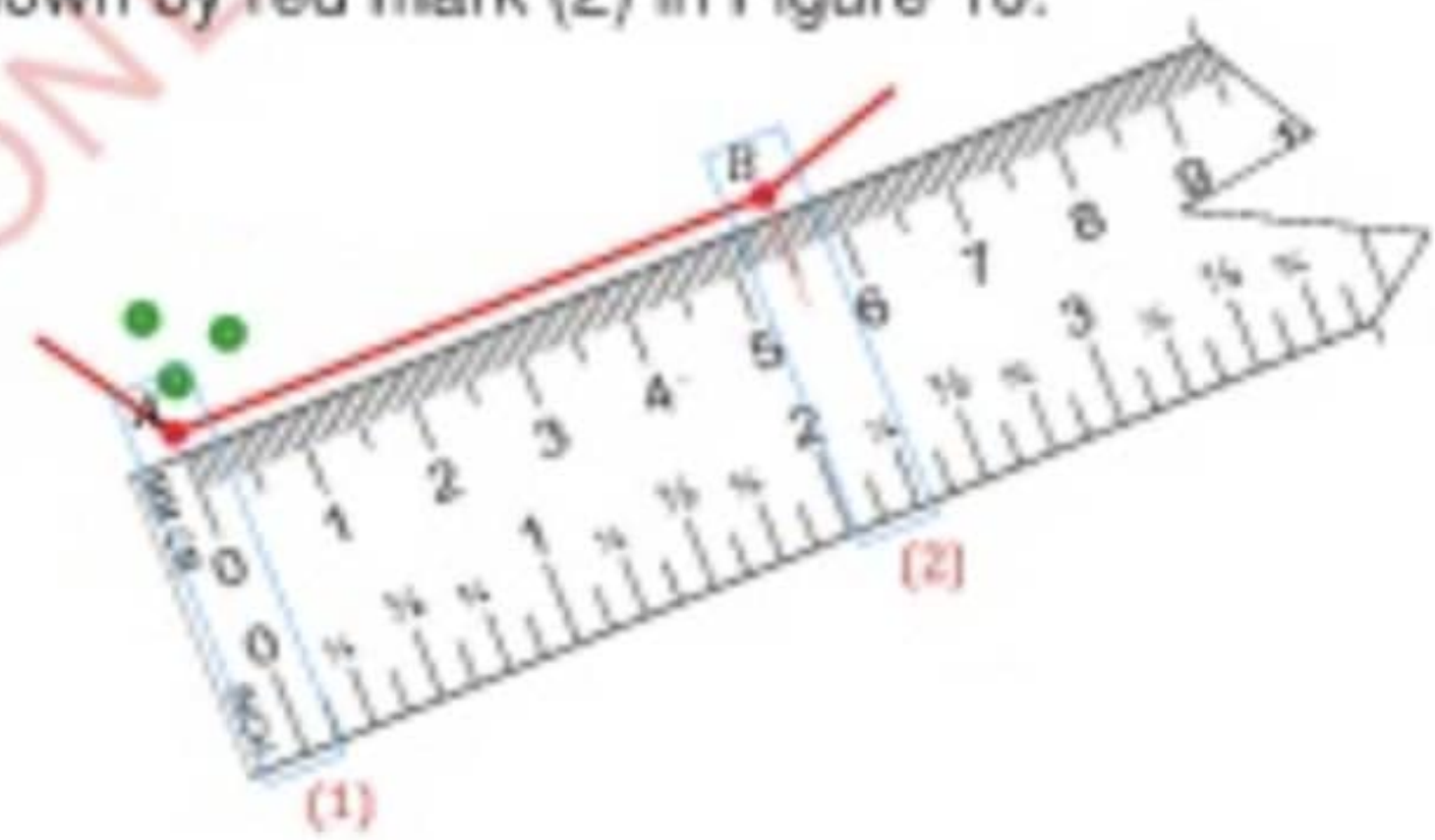


Figure 10: *Measuring straight distance using a ruler*

(c) The map distance measured by a ruler between points A and

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B can be converted into the ground distance using a linear or representative fractional scale, following these steps:

(i) Place a ruler on the linear scale by aligning its zero (0) with a zero (0) on the linear scale, and the red mark on the right side;

(ii) if the red mark is aligned with a certain number, that number on a linear scale represents the ground distance between points A and B;

(iii) If the red mark is between two numbers, record the left number and mark its position on the ruler as shown in Figure 11. For example, the left number in Figure 11 is 2 km, then record it;

(iv) move the ruler to the left of a linear scale to align the right red mark with zero (4) (the left red mark will be on left side (5);

(v) record the closest number to the left red mark. For example, in Figure 11 the recorded number is 750 metres. Therefore, the ground distance from point A to B is 2 km and 750 metres.

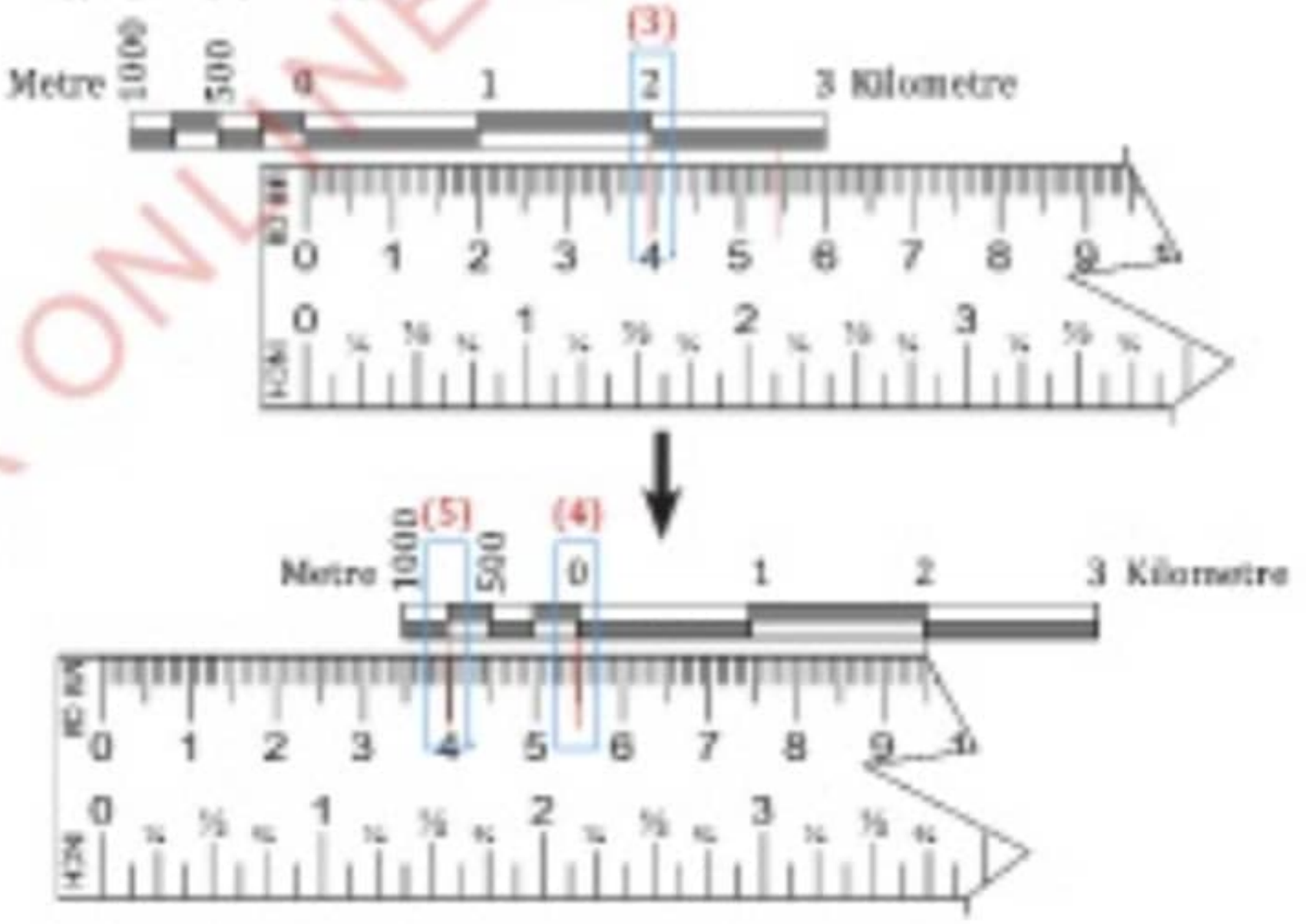


Figure 11: *Converting map distance to ground distance when the right red mark representing point B is between two numbers on a linear scale.*

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(d) By using a representative fractional (ratio) scale. Ground distances from point A to B can be calculated as in straight distances measured using a piece of paper. Since a map distance of 5.5 cm was measured in Figure 10, and the map scale is 1: 50000, then the ground distance will be 2.75 kms.

Activity 5

Observe Figure 12, then answer the following questions:



Figure 12: *A map of Gezaulole*

Questions

By using a ruler;

(a) Calculate the distance from Kitunda Mission to the village office.

(b) Calculate the distance from the village office to the hamlet office.

(c) Calculate the distance from the hamlet office to Kitunda Mission.

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Measurement of curved distance

To measure the distance of curved features such as rivers, roads, or railways, tools like a pair of dividers, a piece of paper, thread or string, and scale are used.

1. *Using thread / string*

The following steps are involved to determine a curved distance on a map using a string:

(a) Identify the two points connected by a curved line. For example, points A and B then divede the distance that needs to be measured into shorter, straight segments and mark them as shown in Figure 13.



Figure 13: The divided segments to be measured from point A to B.

(b) Take the thread and tie a knot at one end on the left side, then lay it along the curved line connecting the two points by tracing the shorter divided segments, and mark the thread towards the other end, tie another knot at the last mark where point B is ended as shown in Figure 14.

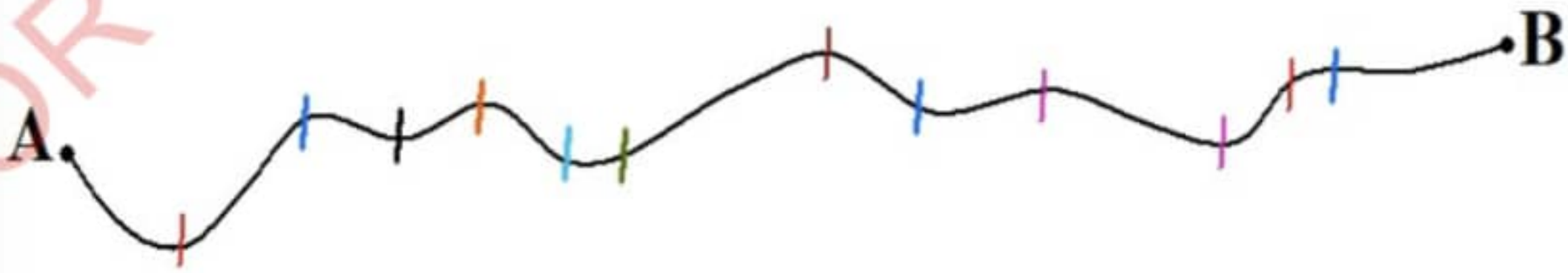


Figure 14: *Tracing the line with a thread from point A to B.*

(c) Use a linear scale to convert map distance to real distance on the ground as follows:

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(i) Stretch the thread and lay it along the linear scale with the first knot on zero (0) and the second knot on the right side of the scale;

(ii) if the second knot aligns with a certain number on the linear scale, that number represents the real ground distance between the two points;

(iii) if the second knot is between two numbers, record the left number and mark its position on the string, as shown in red in Figure 15. For example, the left number recorded in Figure 15 is 1 km. Then, move the thread to the left until the second knot aligned with zero (0) and the marked point is on the left side of the scale; and

(iv) finally, record the number at the linear scale close to where a string red mark anded. For example, 500 metres is recorded as shown in Figure 15. Therefore, the ground distance between points A and B is 1 km and 500 metres.

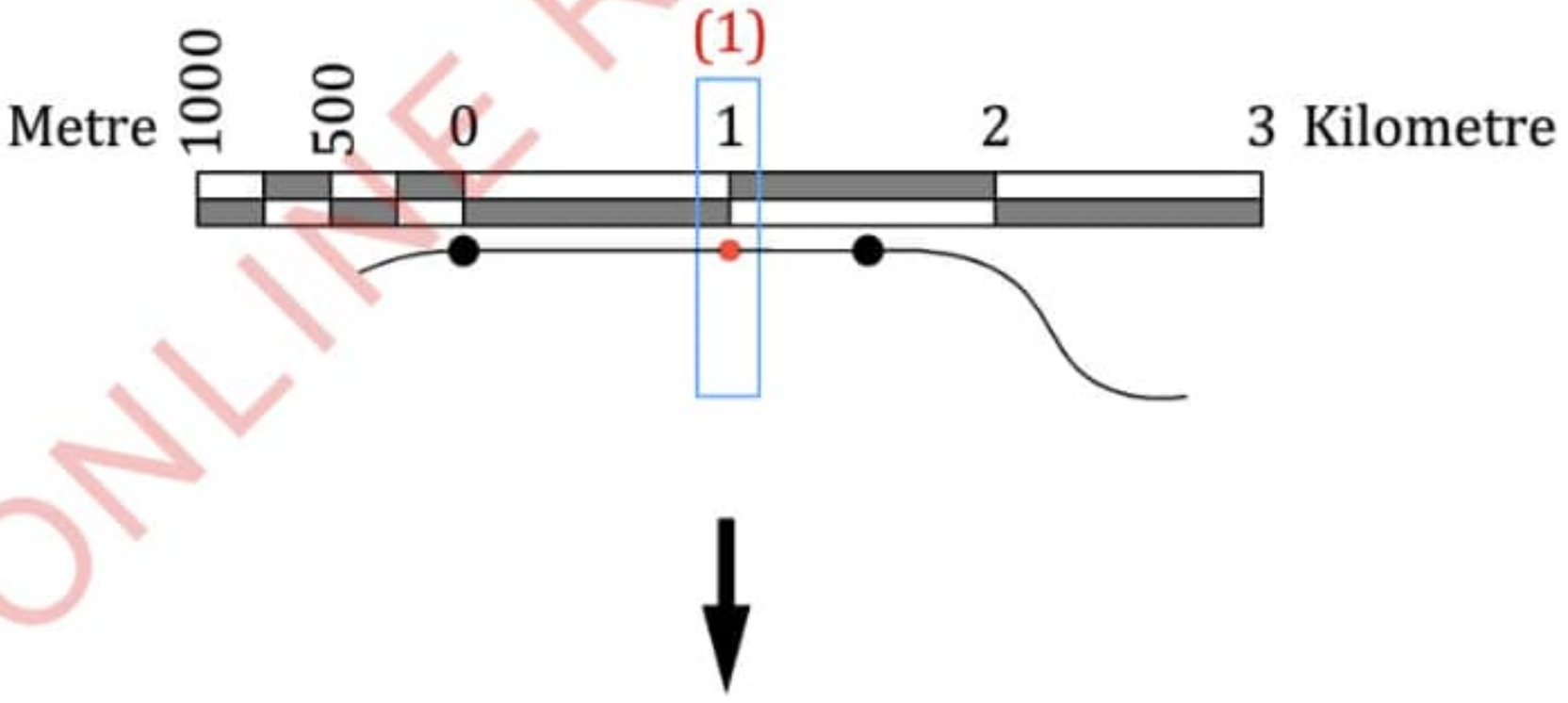


Figure 15: *Converting map distance measured by thread into the ground distance*

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(d) By using a representative fractional (ratio) scale, ground distance from point A to B can be calculated as in the measurement of straight line distance using a piece of paper. If the representative fractional scale is 1: 50000 and a map distance measured between the two knots is 3cm, then the ground distance between points A and B is 1.5 km.

2. *Using a pair of dividers*

The following are the steps to measure distance on a map using a pair of dividers:

(a) Identify two points, namely point A and B as shown in Figure 16.

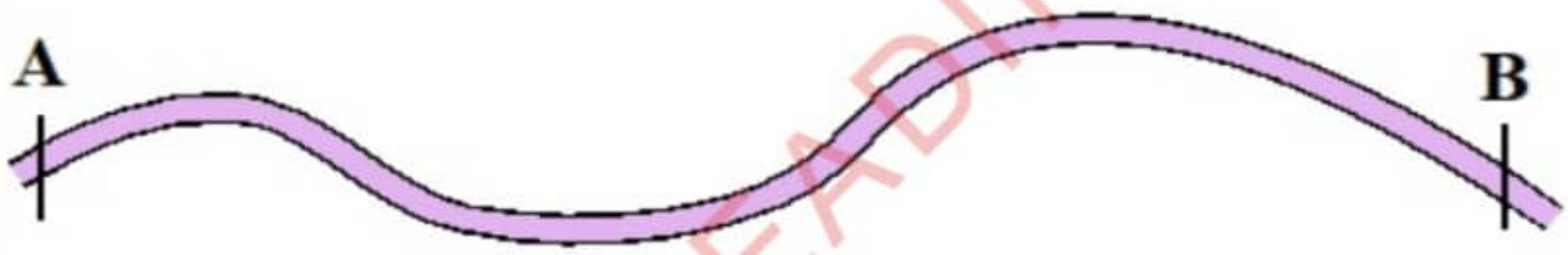


Figure 16: *The curved distance to be measured from point A to B*

(b) Divide the distance that needs to be measured into shorter, straight segments and mark them as shown in Figure 17

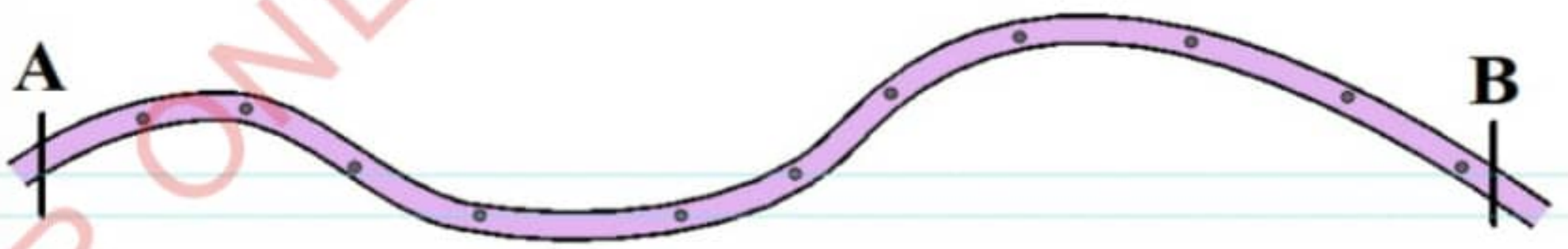


Figure 17: *The divided straight segments to be measured from point A to B*

(c) Use a pair divider to obtain the length of each segment one after the other, as shown in Figure 18

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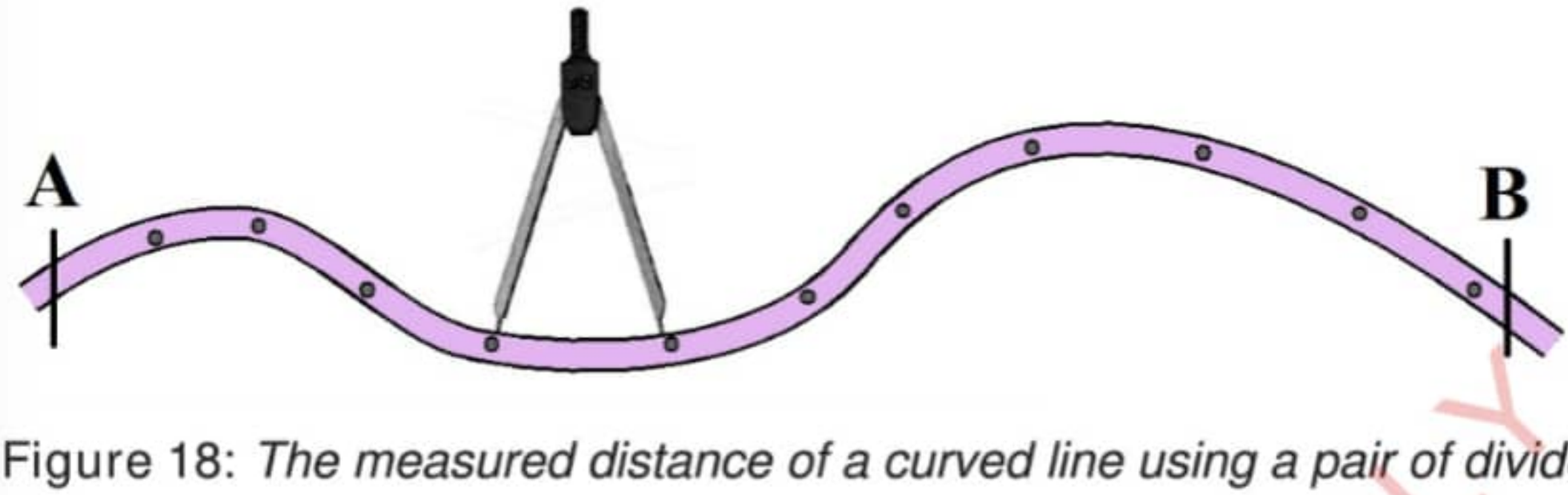


Figure 18: *The measured distance of a curved line using a pair of dividers*

(d) Measure the distance of each segment you divided using a ruler and record them on paper as shown in Figure 19;

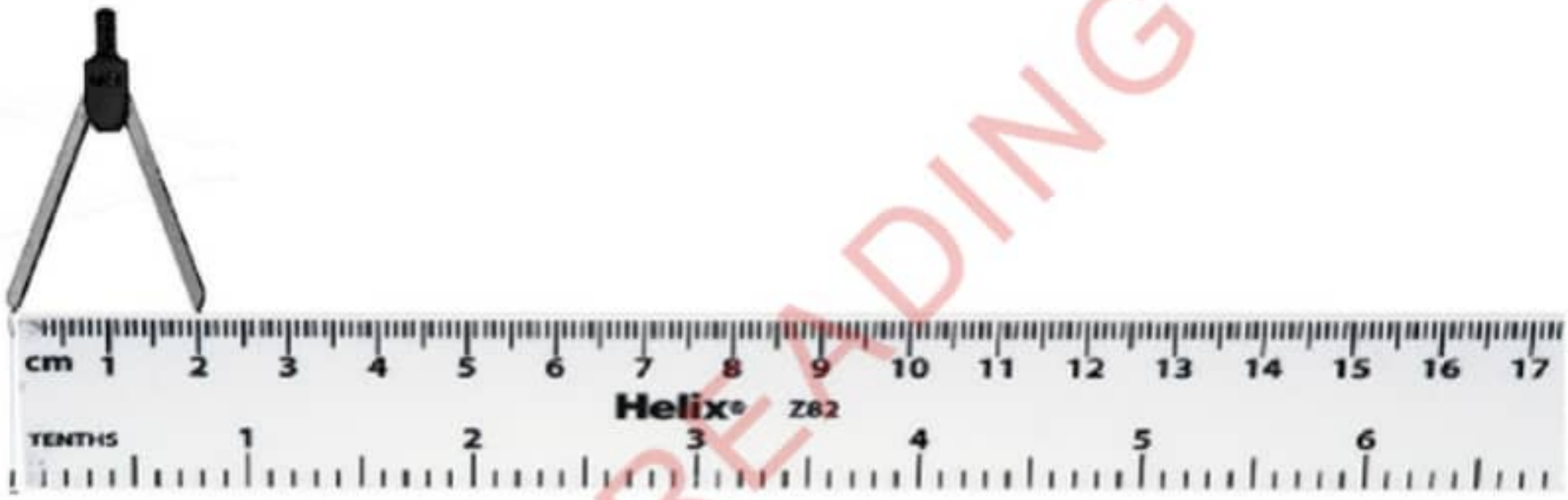


Figure 19: *A ruler and a pair of dividers*

(e) Add the measured distances to obtain the total distance from points A to B; and

(f) Convert the obtained map distance into the ground distance using the representative fractional scale as in the measurement of straight distances using a piece of paper. For example, if a total map distance obtained in step (d) was 14 cm and the representative fractional scale is 1:50000, then the actual ground distance between points A and B is 7 kilometres.

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Activity 6

Observe Figure 20, then answer the following questions:



Figure 20: *A map of Ulole*

Questions

1. Use a thread to calculate the distance from Kilimani to Hekima.
2. Use a piece of paper to calculate the length of the railway.
3. Use a pair of divider to calculate the length of river Bariadi from the mountain peak to Lake Gagadi.

Measuring distance digitally

Apart from determining the distance of a straight and a curved object using traditional methods, we can also determine distances on a map using digital methods. The following steps explain how to use

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"Google My Maps" software to measure distance.

(a) On your computer or smartphone, open the Google browser, search for "Google My Maps", and click on My Maps as shown in Figure 21.

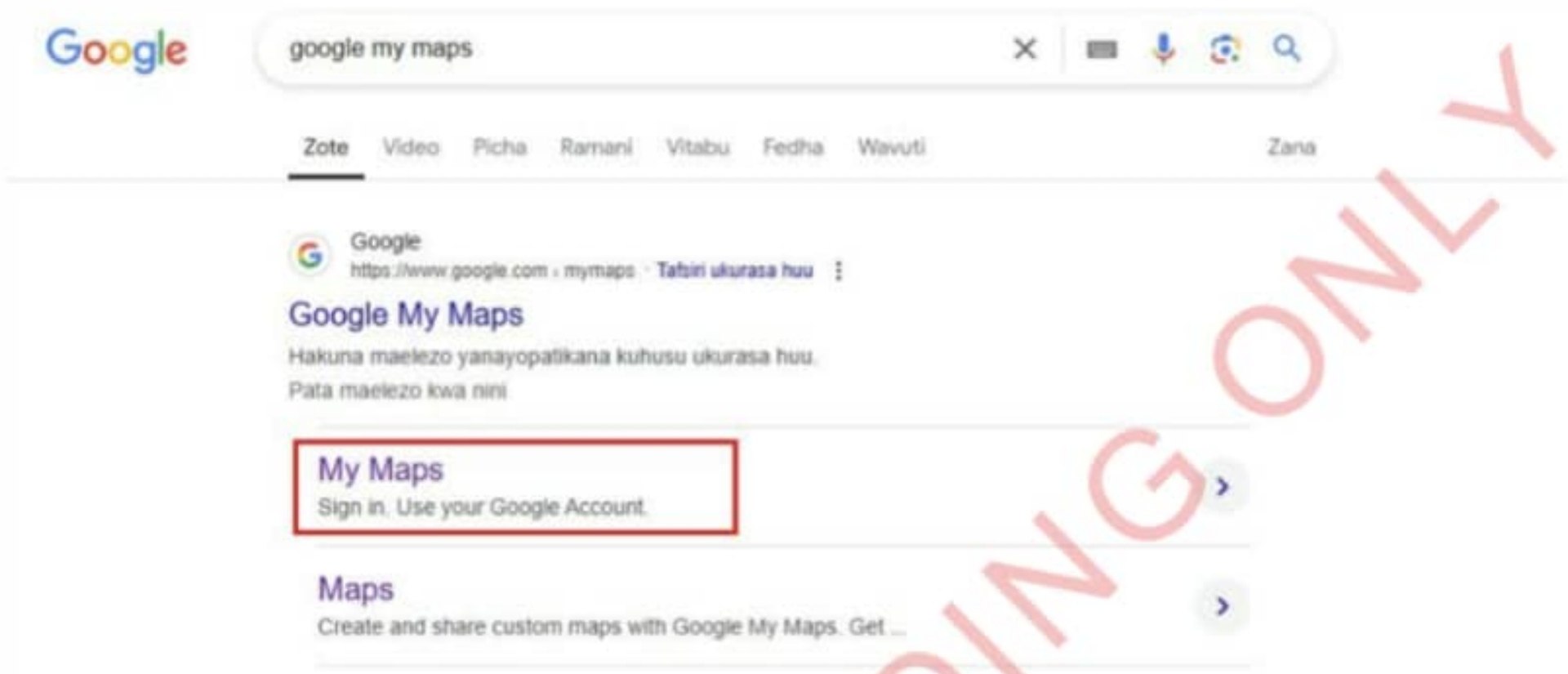


Figure 21: *Opening Google My Map software*

(b) After Google My Maps is opened, search for the area where you are going to measure the distance between two points. For example, we are going to search for Munguri FDC as shown in Figure 22 to measure the distance between Munguri FDC and Munguri Primary School.

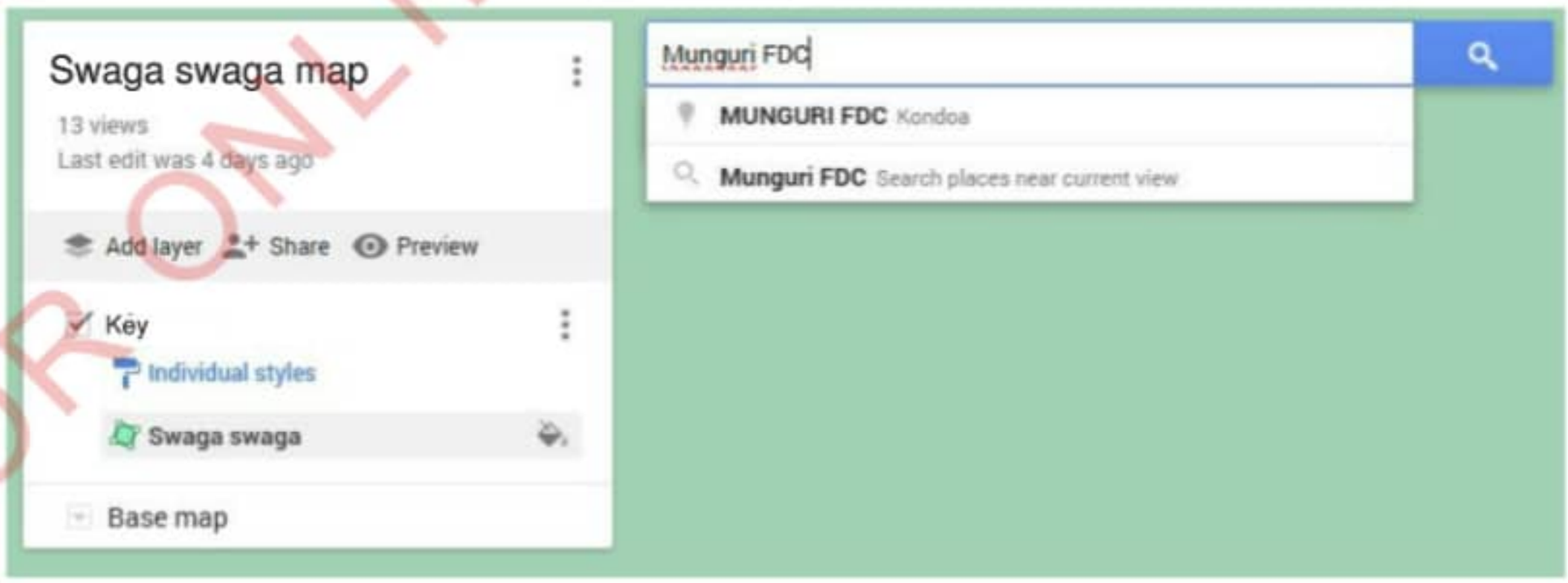


Figure 22: *Search for the area where you want to measure a distance*

(c) Once Munguri FDC is shown on Google My Maps, zoom out by moving the mouse wheel backwards to view Munguri FDC and Munguri Primary School as shown in Figure 23. This will

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show the route to follow when measuring the distance from Munguri FDC to Munguri Primary School.



Figure 23: *Zooming out to show the route connecting the two points*

(d) To measure the distance, select the ruler icon as shown in Figure 24. This icon allows you to measure distances for features like roads, railways and rivers and areas for features like forests, dams, lakes, and administrative boundaries.

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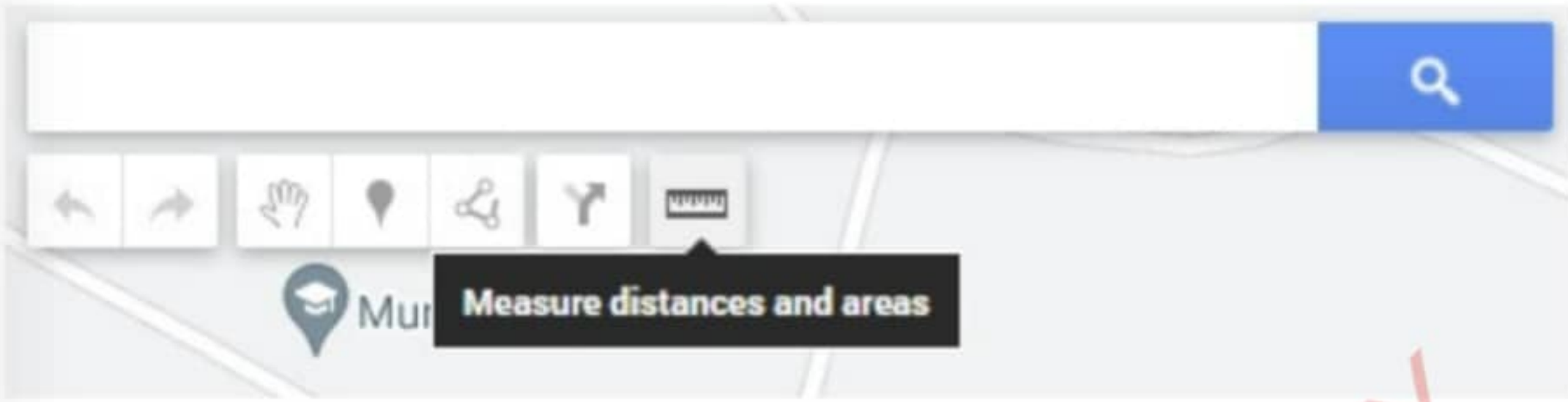


Figure 24: *Selecting the ruler icon for distance measurement*

(e) Then, move the cursor and click on the first point (that is Munguri FDC) (1) and start measuring the distance by tracing the route towards Munguri Primary School as shown in Figure 25. Continue tracing the route until you reach Munguri Primary School. Once you reach Munguri Primary School, double-click to finish measuring the distance and the distance obtained (4) is displayed as shown in Figure 25. Therefore, the distance from Munguri FDC to Munguri Primary School is 1.46 kilometres.

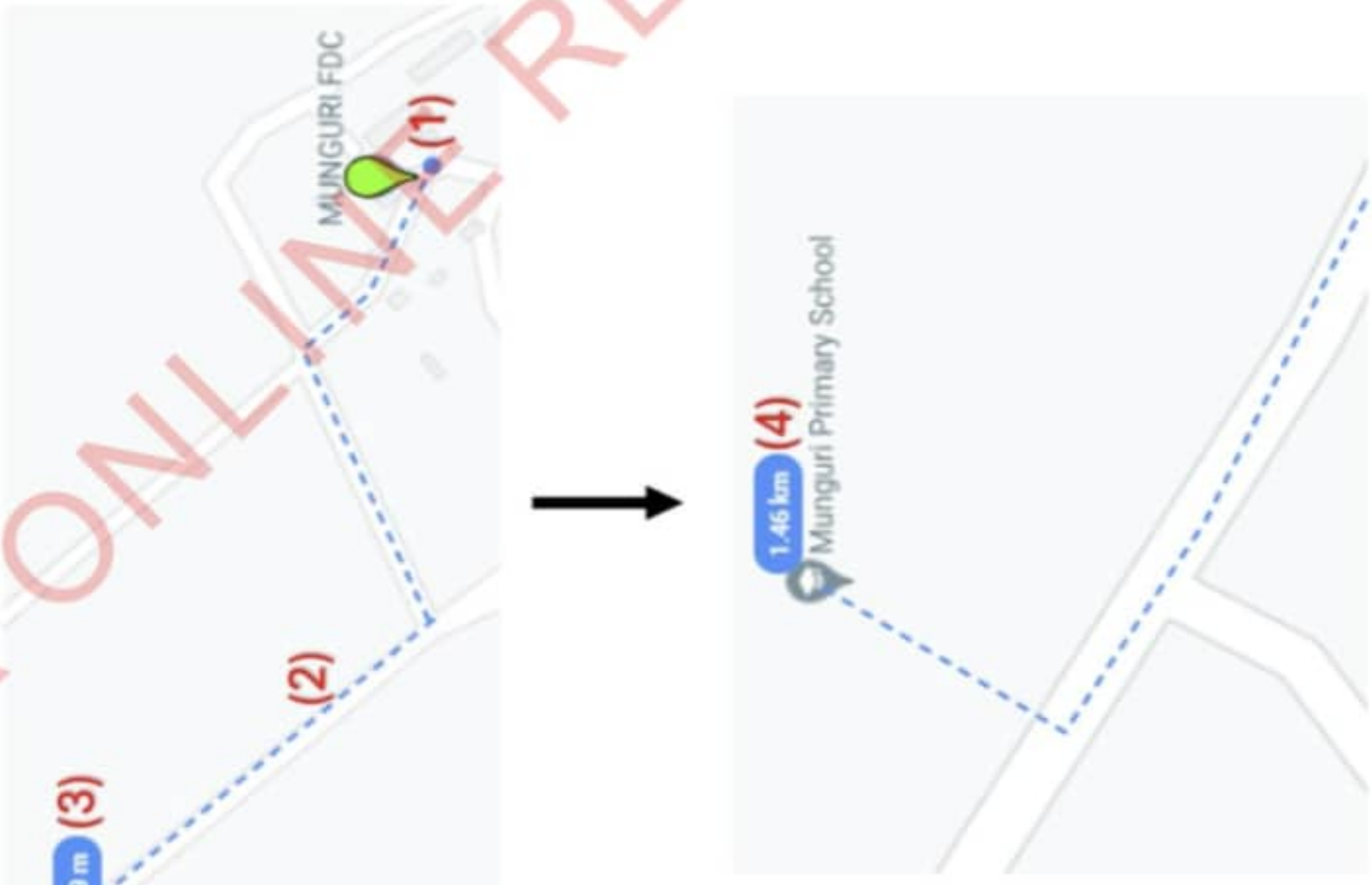


Figure 25: *Tracing the route to measure the distance from Munguri FDC to Munguri Primary School*

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Using a map to identify administrative boundaries

Activity 7

Use online reliable sources or books to study the map of Tanzania and identify different administrative boundaries.

The following are important aspects to consider to identify administrative boundaries correctly:

(a) Selecting the appropriate map;

(b) Identifying a reliable source of information including Google Maps, OpenStreetMap, or special government software; and

(c) Recognizing the symbols and signs used on the map to show administrative boundaries. Several times the lines of different sizes, shapes or colours used to represent different types of administrative boundaries. Some boundaries are identified by various features, such as rivers, lakes, mountains, and roads.

Generally, identifying administrative boundaries is important for various purposes, including governance, decision-making, the management of public resources and the implementation of policies and development plans.

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Activity 8

Observe Figure 26, then answer the following questions:

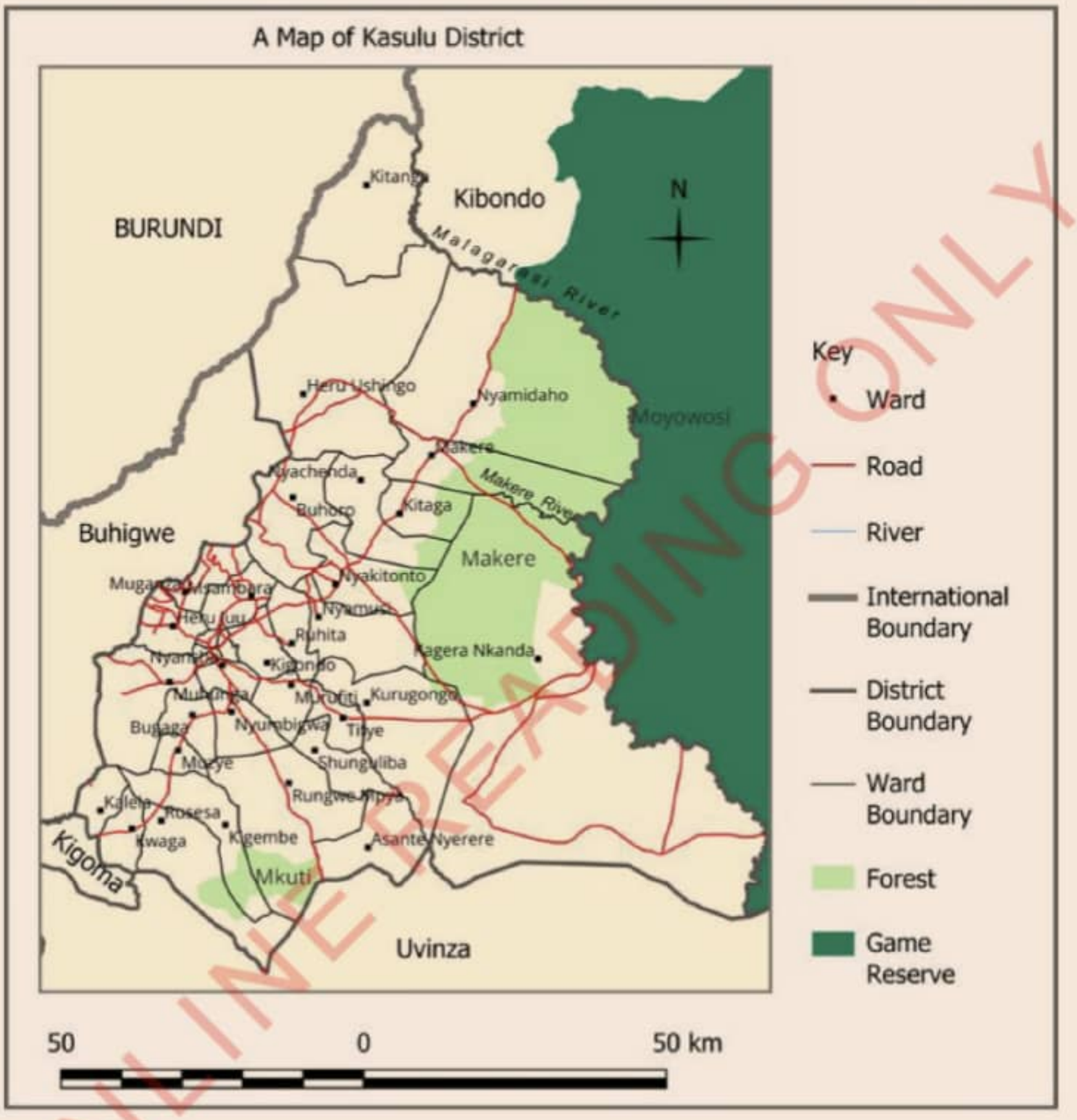


Figure 26: *Kasulu District*

Questions

1. Which districts are bordered kasulu district to the (a) West (b) Northeast (c) Southwest and (d) South.
2. What does Makere forest border on the eastern side?
3. Identify the boundary that separates the Moyowosi game reserve from Makere forest.
4. What does Kitaga ward border on the eastern side?
5. What does Kwaga ward border on the Southeast side?

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Activity 9

Observe Figure 27, then answer the following questions:



Figure 27: *A map of the Mtwara Region*

Questions

1. Which administrative boundaries are identified on the map?
2. How can you differentiate between the regional boundaries and district boundaries as indicated in Figure 27?
3. What are the social and economic effects that can occur if administrative boundaries are not correctly identified?

Using maps to identify Earth's relief

Relief refers to the elevetion of the land including mountains, valley, hills, plains and plateus. Maps use different symbols and colours to represent the Earth's relief. For example, contour lines on a map are used to indicate ground elevation or depression of the land.

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Mountains and hills

Often, mountains and hills are shown with brown or red colours or contour lines. The darker the colour, the higher the mountain. Or, if the contour lines are too close, it means the land is steep, like a mountain. Hills, which are smaller than mountains, are usually indicated with lighter brown or green colours or by close contour lines.

Valleys and plains

Valleys are the low areas between mountains or hills. On a map they can be shown with green colours or inward-curving contour lines. Plains, which are flat areas, are often represented with green colours. Also, they can be represented with contour lines. If the lines are far apart, it means that the land is flat.

Plateaus

These are flat areas on top of high land, usually shown with brown colours. Also on a map, a plateau can be shown by contour lines that are close on the sides and spread out on top. The spread-out lines show the flat part of the plateau.

Generally, when using a map to identify relief of a certain area, pay attention to the colours or contour lines. Understanding the relief of the land is useful for various purposes, such as planning where to build, and learning about the environment.

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Activity 10

Observe Figure 28, then answer the following questions:

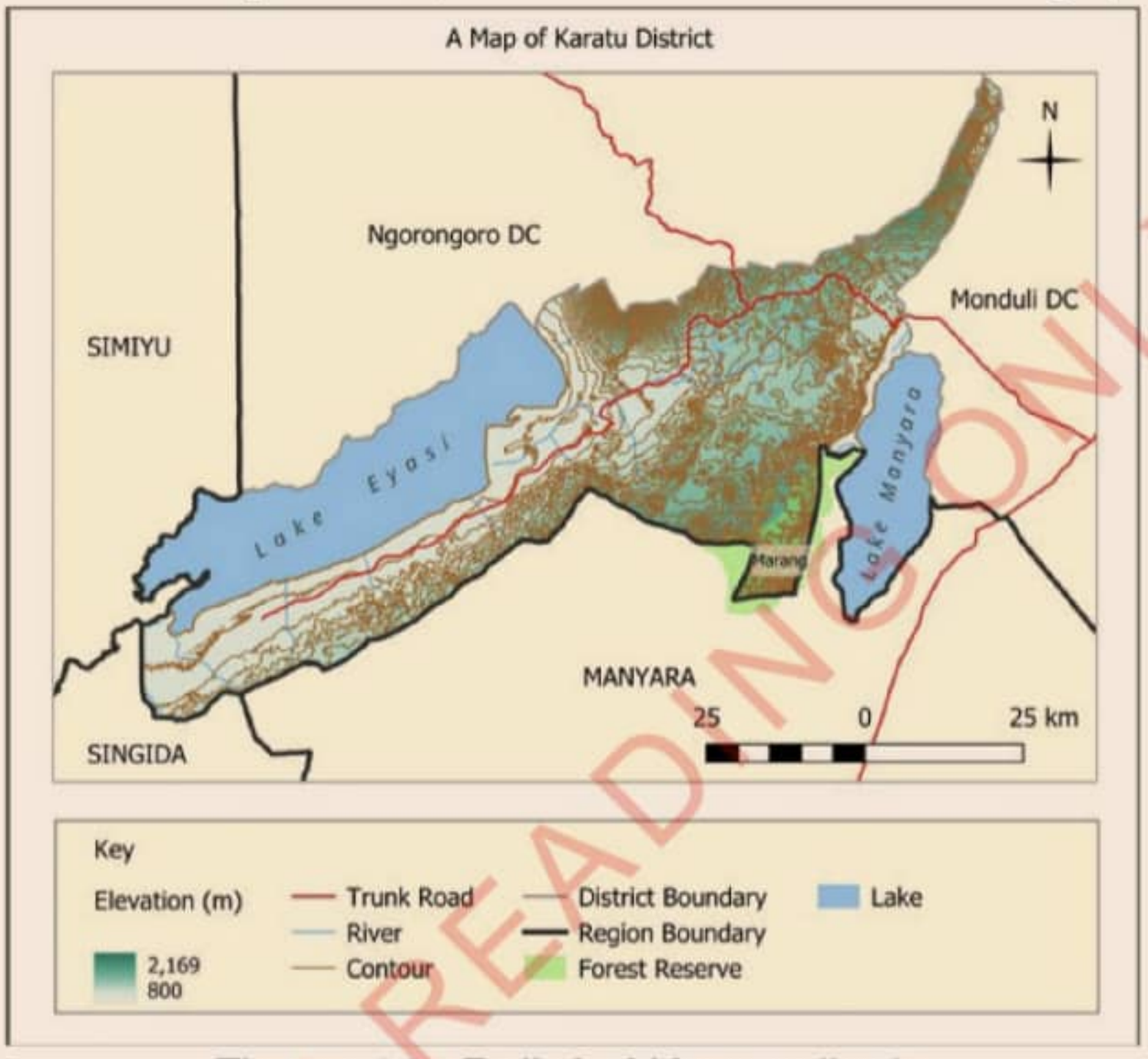


Figure 28: *Relief of Karatu district*

Questions

1. Identify the relief features represented on the map.
2. What economic activities can take place in the relief features you identified?
3. Identify the symbols used on the map to represent the relief features.

Using maps to determine the distribution of features and people

Activity 11

Use books or online reliable sources to learn how the distribution of people and objects are represented on a map.

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Maps are essential tools for identifying where people live and where different features are found. Maps can use dot symbols, or colours to show distribution of people. For example, if you see many dots close together on a map, it means many people live in that area, like in big cities. If the dots are spread out, it means fewer people live there, like in the village or small towns.

The distribution of people and objects also can be determined by using Geographic Information Systems (GIS) and other digital tools. This helps us to understand where people are more or less crowded.

Maps also use specific symbols to represent the distribution of different features on the earth's surface. For example, house symbol shows where homes are located, while a school symbol shows where schools are found. Similarly, maps use different colours to show features. For example, blue might be used for water sources like rivers and lakes, green for areas with forests, and yellow for desert areas.

Generally, you are advised to use the key to see what symbol or colours used to represents distribution of objects and people so as you can use the map properly.

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Activity 12

Observe Figure 29, then answer the following questions:

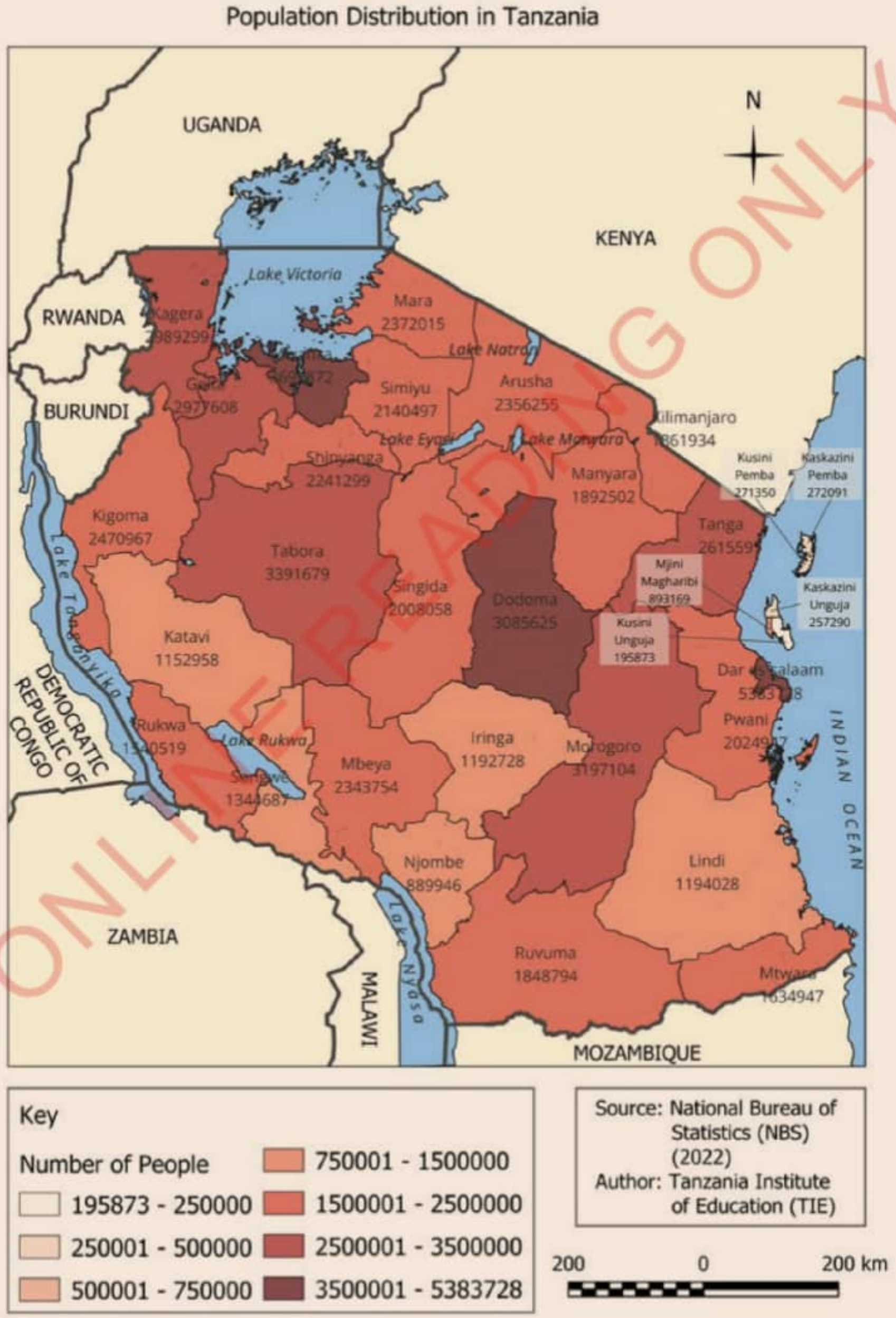


Figure 29: *Population distribution in Tanzania*

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Questions

1. What is the lowest population size?
2. What is the most populated region?

Activity 13: Use Google My Maps software

1. Use a smartphone, tablet or computer to open your browser, and type Google My Maps.
2. Using the Google My Maps interface, click on the search section.
3. Type the name of your district in the search section, then click search.
4. After that, find the number of primary schools available in your district by typing, for example, 'primary schools in Chamwino District; and
5. Describe the distribution of primary schools in your district.

Exercise 1

1. Explain the importance of the map key in identifying the distribution of people and features on a map.
2. What places does low population normally represent on a map?
   (a) Urban areas
   (b) Rural areas
   (c) Near rivers
   (d) Near the ocean

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3. If you see dots close together on the map, what does it mean?
   (a) There are few people
   (b) There are many people
   (c) There are many trees
   (d) There is a lot of water
4. What do scattered dots on a map indicate?
   (a) There are many people
   (b) There are few people
   (c) There are many cars
   (d) There are many schools
5. Which of the following can help you to identify the distribution of people and features?
   (a) A storybook
   (b) A radio
   (c) Geographic Information Systems (GIS)
   (d) A watch

Revision exercise

A: Choose the letter of the correct answer

1. Important things to consider regarding determining the correctness of administrative boundaries of an area on a map include_____
   (a) using a map with detailed information.
   (b) using a map with a reliable source of information.
   (c) using a map with acceptable colours.
   (d) using a map that shows the elevation.
2. On a map, administrative boundaries can be identified using:

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(a) key, title, and source.

(b) symbols, colours, and Geographic Information System (GIS) technology.

(c) special software.

(d) compass and scale.

3. If the map distance from Point A to B is 50 cm and the scale is 1:100000, what is the actual ground distance in kilometres?

(a) 5 kilometres.

(b) 25 kilometres.

(c) 50 kilometres.

(d) 100 kilometres

4. Valleys on the map are shown using _____.

(a) curved inward contour lines

(b) closest contour lines

(c) dispersed contour lines

(d) straight contour lines

5. What does the light blue colour usually represent on a map?

(a) Water sources like rivers and lakes

(b) Forests

(c) Mountains

(d) Buildings

Section B. Matching items

6. Match the items in column A with those in column B to produce a meaningful statement

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No.	Column A	Column B
(i)	A symbol of a North direction	(a) help to make correct decisions for the management of the country's resources.
(ii)	Contour	(b) affects the lives of living organisms.
(iii)	Dots	(c) used to identify the shortage of inputs in the country.
(iv)	GIS	(d) used to show steep and gentle slopes on a map.
(v)	Administrative boundaries	(e) used to show population distribution.
		(f) determine the direction and location on a map.
		(g) Geographical Information System.

Section C: Short answer questions

7. Explain how contour lines appears on a map to show the following features on a map.
   (a) Steep slope;
   (b) Plain; and
   (c) Valley;
8. Explain how you can use a map to plan a trip to visit a national park.
9. How can you identify the direction by using a map?
10. How can you determine the actual distance by using a map?

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Vocabulary

Contour	Is a line drawn on a map to connects points of equal elevation above the sea levels
GPS:	Is a Satellite system used to determine the position of an object on the Earth's surface

Test

Section A: Choose the letter of the correct answer

1. What is the main purpose of a map?
   (a) To show how to make a drawing.
   (b) To show the location of features on the Earth's surface.
   (c) To show how to calculate distances.
   (d) To show rescue methods.
2. Before choosing a map for your needs, what is an important question to ask yourself?
   (a) What scale does that map use?
   (b) What is the purpose of the map?
   (c) What colour is the map?
   (d) How large is the map?
3. Which essential element of a map can be used to determine the actual distance on the ground?
   (a) Key.
   (b) Scale.
   (c) Title.

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(d) Gridlines.

4. Which characteristic of a map ensures that the information shown is accurate?

(a) It should use clear colours and neat writing.

(b) It should have the up to date information.

(c) It should be easy to read and interpret.

(d) It should indicate the title of a map.

5. Why is it important to use symbols, colours, and styles consistently on a map?

(a) To make the map more colourful.

(b) To help users understand it better.

(c) To show more information.

(d) To increase the map's size.

6. How is the descriptive scale presented on a map?

(a) By using numbers to show the ratio.

(b) By using a line to indicate distance.

(c) By using words to describe the ratio.

(d) By using a compass.

7. What does a scale represented as 1:50000 indicate?

(a) 1 centimetre on the map represent 50000 centimetres on the ground.

(b) 1 meter on the map represents 50000 meters on the ground.

(c) 1 unit on the map represents 50000 units on the ground.

(d) 1 kilometre on the map represents 50000 kilometres on the ground.

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8. Why is it important to know Cardinal Directions?

(a) To know how to draw maps.

(b) To find direction and location.

(c) To understand weather systems.

(d) To understand features on the Earth's surface.

9. What type of scale is used to convert map distances into actual ground distances?

(a) Linear scale.

(b) Temperature scale.

(c) Time scale.

(d) Distance scale.

10. The North arrow is used to identify which direction on a map?

(a) East

(b) South

(c) North

(d) West

11. Which software can be used to measure distances on a map?

(a) Google My Maps.

(b) Calculator.

(c) Stopwatch.

(d) Thermometer.

12. On a map with a scale of 1:50000, how many kilometres on the ground does 2 centimetres on the map represent?

(a) 1 kilometre.

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(b) 2 kilometres.

(c) 5 kilometres.

(d) 10 kilometres.

Section B: Short answers questions

13. What is the first step to measuring straight distance on a map using a piece of paper?
14. What symbol is commonly used to represent elevation areas on a map?
15. Which symbol on a map is often used to show administrative boundaries?
16. Which methods can be used to measure the curved distance on a map?
17. Describe things you should look for on a map to help you find your way to a place you have never been before.

Section C: Matching items

18. Match the items in column A with those in column B to produce a meaningful statement

No.	Column A	Column B
(i)	Cartography	(a) shadow falls to the West.
(ii)	Helps in identifying directions on maps	(b) shows countries, regions, and districts with boundaries.
(iii)	Map symbols	(c) distances on a map and actual distances on the ground.
(iv)	Digital map drawing	(d) helps to find routes for travel.
(v)	Key	(e) shows elements of weather like rain, and temperature

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(vi)	Simple map	(f) the drawing of maps using computer softwares.
(vii)	Political Map	(g) study about map drawing.
(viii)	Weather map	(h) Symbols used on maps to represent various features like roads, buildings, and water bodies.
(ix)	Road map	(i) simple drawing of an area that drawn without precise measurements.
(x)	when walking towards the Sun in the morning	(j) used to explain different symbols used on a map.
		(k) shadow falls to the East.
		(l) North direction.

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Section D: Map reading

Observe the following map then answer the questions that follow:

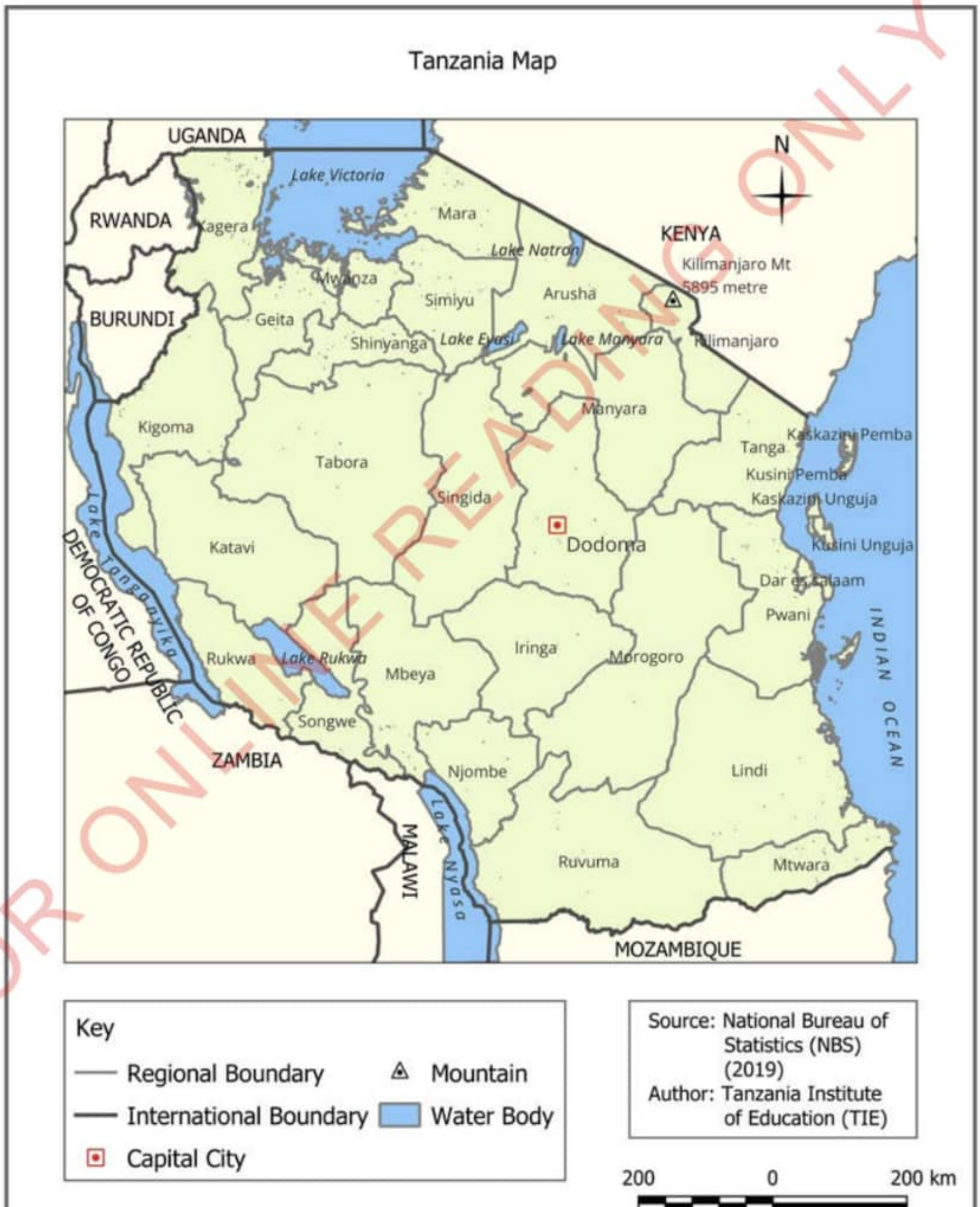


Figure 30: *Map of Tanzania showing administrative boundaries*

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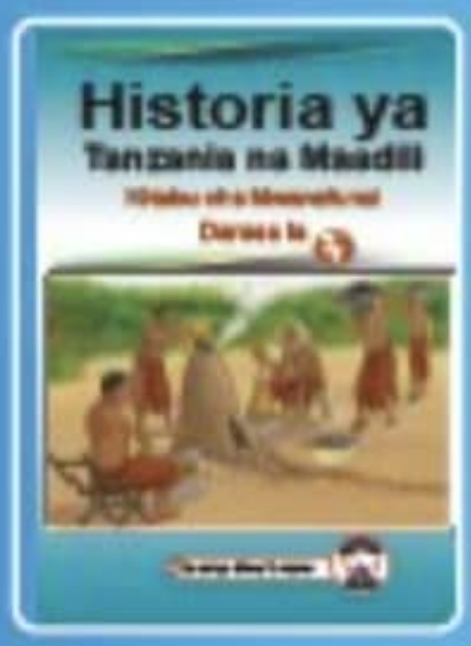
19. Which side of Tanzania is the Mtwara Region located?
20. Name the regions bordering the Dodoma Region to the East, West, South, and North.
21. Which regions border Lake Tanganyika?
22. Which lake is located in the southern part of Tanzania?
23. In which direction is Lake Victoria located in Tanzania?
24. Where is the capital city of Tanzania?
25. What does Tanzania border on the eastern side?

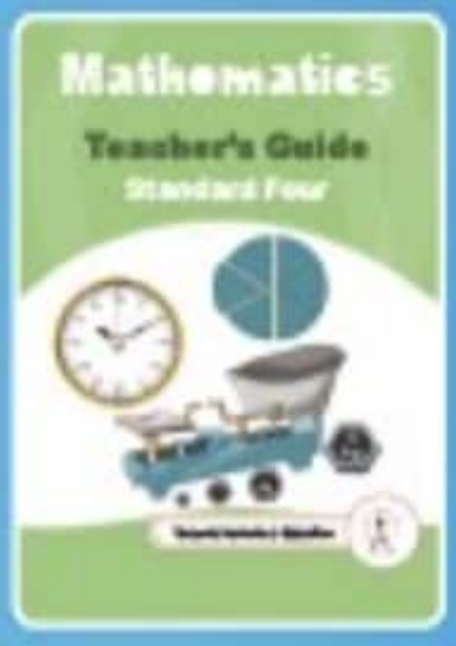
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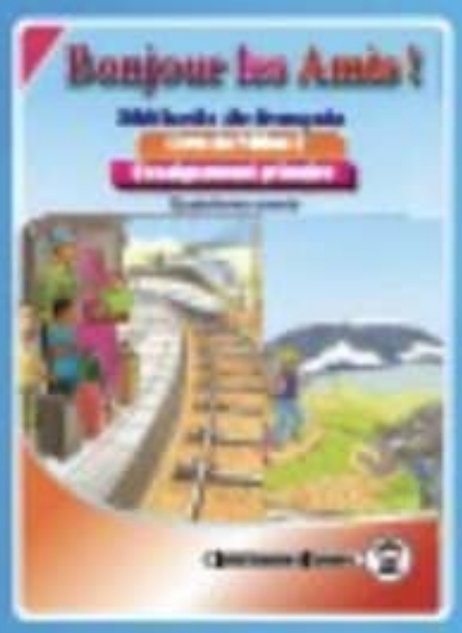
Other books from Tanzania Institute of Education

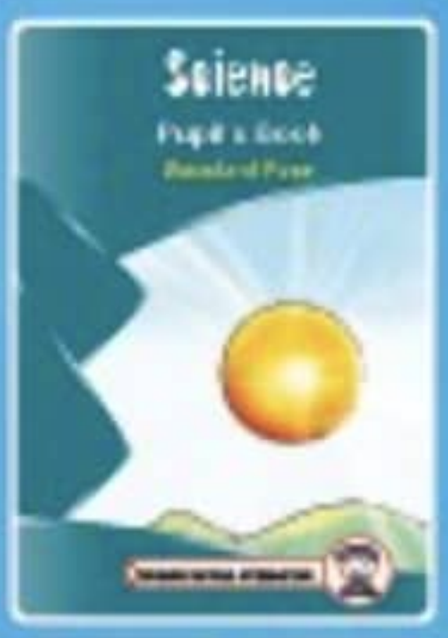
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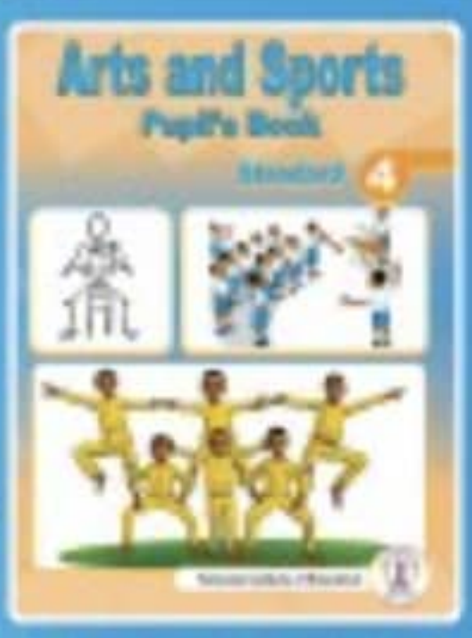


















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